MUSCLE OS

PILLAR 6

Hormonal Optimization

the complete guide to optimize hormones for body composition and performance

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DOMAIN 1

Hormonal Foundations: the endocrine system, its axes, and the core hormones that govern body composition

1. The Endocrine System and Body Composition

1.1 The Endocrine System as the Master Regulator

The endocrine system is the bodys network of glands and organs that synthesise and secrete hormones, chemical messengers that travel through the bloodstream to exert effects on distant target tissues. In the context of body composition and athletic performance, hormones determine whether ingested nutrients are stored as fat or synthesised into muscle, whether training stimuli produce adaptation or burnout, and whether recovery restores homeostasis or accumulates deficit. The endocrine system is the rate-limiting step

1.2 Hormone Classification

Table 1.1 Hormone Classification by Chemical Structure

CLASS	EXAMPLES	SOLUBILITY	MECHANISM OF ACTION	ONSET
Peptide/protein hormones	Insulin, GH, IGF-1, leptin, prolactin	Water-soluble	Bind membrane receptors; second messenger cascades	Minutes
Steroid hormones	Testosterone, estrogen, cortisol, progesterone	Lipid-soluble	Diffuse through membrane; bind intracellular receptors; alter gene transcription	Hours to days
Amino acid derivatives	Thyroid hormones (T3, T4), catecholamines (adrenaline)	Variable	T3/T4 bind nuclear receptors; catecholamines use GPCRs	Minutes to hours

1.3 Hormonal Impact on Body Composition

Each major hormone exerts a specific signature on body composition. Testosterone promotes muscle protein synthesis (MPS), satellite cell activation, and inhibits adipogenesis. Cortisol promotes proteolysis, gluconeogenesis, and visceral fat accumulation. Thyroid hormones set the basal metabolic rate and influence lipid oxidation. Insulin is the master anabolic storage hormone, driving nutrients into cells. Growth hormone and IGF-1 mediate longitudinal bone growth, lean mass accretion, and lipolysis. Estrogen influences fat distribution, bone mineral density, and insulin sensitivity. The net body composition phenotype is the sum of these hormonal inputs, modulated by receptor sensitivity and binding protein concentrations.

> **Hormonal Milieu Defined**
>
> **Hormonal milieu**: The integrated hormonal environment in which all physiological processes occur. It is the net effect of hormone concentrations, binding protein levels, receptor density, receptor sensitivity, and intracellular signalling efficiency. Coaching interventions aim to optimise the milieu, not manipulate individual hormones in isolation.

1.4 The Principle of Hormonal Primacy

When designing interventions for body composition, the hormonal environment takes precedence over isolated nutritional or training variables within physiological ranges. A client with suboptimal testosterone, elevated cortisol, or disrupted thyroid function will not respond predictably to standard protocols. The coaching hierarchy must therefore begin with hormonal assessment as a prerequisite to programme design. This does not mean every client needs blood work; it means every coach must understand the hormonal signatures of common presentations (low energy, poor recovery, unexplained fat gain) and know when to refer for clinical evaluation.

CHAPTER SUMMARY

The endocrine system is the master regulator of body composition and performance. Hormones are classified as peptide, steroid, or amino-acid derivatives, each with distinct mechanisms and time courses. The hormonal milieu, the integrated net effect of all hormones, receptors, and binding proteins, determines the outcome of any training or nutritional intervention. Coaching interventions should be designed with hormonal primacy in mind: assess the hormonal environment before prescribing protocols.

◈ MAKE THIS WORK FOR YOU

The hormonal environment determines how your body responds to training and nutrition — this is the "assess before you prescribe" chapter. Most people do not need blood work before starting a programme; they need to know **which symptoms point to a hormonal problem** versus a normal training issue.

If you are a coach or self-coached lifter: The coaching hierarchy is: screen for hormonal red flags before writing protocols. Red flags that warrant blood work before (or alongside) programme design: unexplained strength/muscle loss despite consistent training, libido loss + fatigue + mood changes together, weight gain that resists every diet, persistent poor recovery with normal training load, and irregular menstrual cycles (females). None of these is a single-day event — all are 4+ week patterns.

If you have no red flags: Do not let the hormonal topic frighten you into expensive testing — start with the basics (training, protein, sleep, stress — Chapters 13–16) and only test when you need an answer. The overwhelming majority of "low T feeling" cases in healthy young men resolve with sleep, weight loss, and stress management before any hormone is measured (Chapter 21).

If you have red flags: Do not self-diagnose from internet lists — get a proper panel (Chapter 20 explains exactly which tests), and work with a clinician. Hormone problems are highly treatable when found early; they are also easy to misattribute. Common masquerades: low thyroid masquerades as "slow metabolism", high cortisol as "lazy", low testosterone as "overtraining". A 30-minute blood test beats 6 months of guessing.

The key mindset: Hormones are a system you can influence with lifestyle (sleep, training, nutrition, stress) — not a fate. The next 24 chapters are organised as: understand (Ch1–12) → modulate (Ch13–19) → apply and test (Ch20–25). Read this book in that order, and you will know exactly when to act and when to refer.

2. The Hypothalamic-Pituitary Axis and Feedback Loops

2.1 The Master Gland Hierarchy

The hypothalamic-pituitary axis (HPA) is the central command centre of the endocrine system. The hypothalamus, located at the base of the brain, receives neural input from the cortex, limbic system (emotion), brainstem (autonomic), and circumventricular organs (blood-borne signals). It integrates these inputs and secretes releasing hormones into the hypothalamic-pituitary portal system, which travel directly to the anterior pituitary gland. The anterior pituitary then secretes tropic hormones that stimulate peripheral endocrine glands to produce their respective hormones.

2.2 The HPA Cascade

Table 2.1 Hypothalamic-Pituitary-Target Gland Axes

HYPOTHALAMIC HORMONE	PITUITARY HORMONE	TARGET GLAND	PERIPHERAL HORMONE	PRIMARY FUNCTION
GnRH	LH, FSH	Gonads	Testosterone, estrogen, progesterone	Reproduction, libido, anabolism
CRH	ACTH	Adrenal cortex	Cortisol	Stress response, metabolism, immune suppression
TRH	TSH	Thyroid	T3, T4	Metabolic rate, thermogenesis, growth
GHRH (+ somatostatin -)	GH	Liver, tissues	IGF-1, direct GH effects	Growth, lipolysis, anabolism
Dopamine (PIF)	Prolactin inhibition	Mammary tissue	Prolactin (inhibited by dopamine)	Lactation, immune modulation, libido suppression

2.3 Negative Feedback Loops

Each HPA axis is regulated by negative feedback: the peripheral hormone feeds back to the hypothalamus and pituitary to suppress further release. For example, elevated cortisol inhibits CRH and ACTH secretion, forming a closed-loop regulatory system. This feedback can be overridden by sustained stress (chronically elevated CRH leads to cortisol receptor downregulation and feedback resistance), which is the endocrine basis of HPA axis dysregulation. Understanding feedback dynamics is essential for interpreting labs: a high TSH with normal T4 suggests subclinical hypothyroidism (the pituitary is shouting at a sluggish thyroid); a low or inappropriately normal TSH with low T4 suggests central (pituitary) hypothyroidism.

Coaching Relevance

When a client presents with low testosterone, the question is not simply how do I raise testosterone? It is **why is the HPG axis suppressed?** Is the hypothalamus not secreting GnRH (functional hypogonadism from stress, low energy availability, sleep deprivation)? Is the pituitary not responding? Are the testes failing (primary hypogonadism)? The answer determines the intervention. HPA axis logic is the foundation of hormonal problem-solving.

2.4 Pulsatility and Rhythms

Most HPA axes release hormones in a pulsatile fashion, not continuously. GnRH must be released in pulses every 60 to 120 minutes to maintain LH and FSH secretion; continuous GnRH infusion paradoxically suppresses the gonadotropins (the basis of GnRH agonist therapy). Cortisol follows a circadian rhythm: peak at 6 to 8 AM (the cortisol awakening response), declining through the day to a nadir at midnight. GH is secreted in pulses primarily during slow-wave sleep. Disruption of these rhythms, through shift work, poor sleep, or chronic stress, is itself a hormonal insult, independent of total daily hormone production.

CHAPTER SUMMARY

The hypothalamic-pituitary axis is the central command of the endocrine system, operating through five major axes: HPG (reproduction), HPA (stress), HPT (metabolism), GH/IGF-1 (growth), and prolactin (lactation/inhibition). Negative feedback loops regulate each axis and can become dysregulated under chronic stress. Pulsatile release is essential for normal function; disruption of hormonal rhythms is itself pathological. Coaching application: understanding the axis hierarchy allows the coach to identify the likely source of hormonal dysfunction and intervene at the correct level.

◈ MAKE THIS WORK FOR YOU

The axis model gives you a diagnostic map: when something is off, the question is always **which level** — the brain (central), the gland, or the target tissue. This determines whether lifestyle fixes will work or whether you need clinical help.

If you are a coach or self-coached athlete: Use the axis hierarchy as a triage tool: (1) HPG (reproduction) — low libido, low energy, menstrual irregularities; (2) HPA (stress) — fatigue, poor sleep, cravings, high resting heart rate; (3) HPT (metabolism) — cold hands/feet, weight gain, constipation, slow heart rate; (4) GH/IGF-1 — poor recovery, subpar muscle growth; (5) prolactin — libido loss + nipple discharge (needs medical attention). Most lifestyle-modifiable problems live on the HPA and HPG axes — which is why sleep and energy availability (Chapters 13–16) are the foundational interventions.

If you are a stressed or overreaching athlete: Chronic stress presses on the HPA axis, which then suppresses the HPG axis (the body prioritises survival over reproduction — this is exactly how overtraining kills libido and menstrual cycles). The practical takeaway: when training performance and libido both drop simultaneously, the diagnosis is usually allostatic load, not a training problem — cut volume, fix sleep, and the axis recovers (Chapter 16).

If you are a beginner or healthy person: You do not need to memorise the five axes — you need the one sentence: your glands follow your brain, your brain follows your lifestyle (light, sleep, food, stress, training). Fix the inputs and the outputs follow. Only when symptoms persist despite clean inputs do you investigate gland-level problems with blood work (Chapter 20).

3. Testosterone: Synthesis, Regulation and Optimization

3.1 The Testosterone Molecule

Testosterone is the primary male sex hormone (androgen) and a critical anabolic hormone for both sexes. It is a C19 steroid hormone synthesised from cholesterol through a series of enzymatic reactions in the Leydig cells of the testes (males), the theca interna cells of the ovaries (females), and the zona reticularis of the adrenal cortex (both sexes). The rate-limiting step in testosterone synthesis is the transport of cholesterol into the mitochondria via the steroidogenic acute regulatory (StAR) protein, which is activated by LH binding to Leydig cell LH receptors.

3.2 The Steroidogenic Pathway

Table 3.1 The Steroidogenic Pathway from Cholesterol to Testosterone

STEP	ENZYME	SUBSTRATE	PRODUCT	LOCATION
1	CYP11A1 (P450scc)	Cholesterol	Pregnenolone	Mitochondria
2	3β-HSD	Pregnenolone	Progesterone	Smooth ER
3	CYP17A1 (17α-hydroxylase)	Progesterone	17-OH progesterone	Smooth ER
4	CYP17A1 (17,20-lyase)	17-OH progesterone	Androstenedione	Smooth ER
5	17β-HSD	Androstenedione	Testosterone	Smooth ER

3.3 Regulation of Testosterone Production

Testosterone synthesis is regulated by the HPG axis: hypothalamic GnRH stimulates pituitary LH secretion, which binds Leydig cell LH receptors to activate the cAMP/PKA signalling cascade and upregulate StAR protein expression. The system is controlled by negative feedback: testosterone (and its metabolite estradiol) feed back at the hypothalamus and pituitary to suppress GnRH and LH secretion. SHBG-bound testosterone does not participate in feedback; only free and albumin-bound testosterone is bioactive and available for feedback regulation. Factors that suppress the HPG axis include: low energy availability (LEA), high cortisol (CRH suppresses GnRH), sleep deprivation, opioids, and excessive aromatase activity (estradiol feedback is more potent than testosterone feedback).

3.4 Testosterone Reference Ranges

Table 3.2 Testosterone Reference Ranges

PARAMETER	MALE (ADULT)	FEMALE (ADULT)	OPTIMAL RANGE (MALE)*
Total testosterone	250 to 1,100 ng/dL	15 to 70 ng/dL	500 to 900 ng/dL
Free testosterone	35 to 155 pg/mL	1 to 5 pg/mL	80 to 150 pg/mL
Bioavailable testosterone	130 to 400 ng/dL	2 to 15 ng/dL	200 to 400 ng/dL
SHBG	10 to 57 nmol/L	18 to 144 nmol/L	20 to 40 nmol/L

∗Optimal ranges are not clinical thresholds; they represent targets for physiological function and performance. Clinical deficiency is defined by the reference lab; but optimal function typically requires levels in the upper half of the reference range for the individuals age.

3.5 Lifestyle Optimization of Testosterone

Within physiological ranges, testosterone can be optimised through lifestyle modification. The hierarchy of evidence-based interventions is: (1) Energy availability: chronic caloric deficit suppresses the HPG axis; the threshold for suppression varies individually but is predictable below 30 kcal/kg FFM/day. (2) Dietary fat: steroidogenesis requires cholesterol substrate; minimum 0.5 to 0.8 g/kg/day; very-low-fat diets (below 20% calories from fat) consistently reduce testosterone. (3) Sleep: LH pulse frequency is highest during REM sleep; less than 5 hours per night reduces testosterone by 10 to 15%; 7 to 9 hours is optimal. (4) Body fat: adipose tissue expresses aromatase, converting testosterone to estradiol; visceral adiposity creates a self-reinforcing cycle of testosterone suppression. (5) Training: resistance training acutely elevates testosterone post-exercise; chronic overtraining without sufficient recovery suppresses it. (6) Micronutrients: zinc, magnesium, and vitamin D are co-factors in steroidogenesis; deficiency impairs production.

Testosterone Optimization Decision Process

1. **Rule out pathology:** If total testosterone below 300 ng/dL with symptoms, refer for clinical evaluation. Test LH, FSH, prolactin, ferritin, TSH, SHBG to identify the level of dysfunction.

2. **Assess energy availability:** Is the client in a caloric deficit? Has it been prolonged (beyond 12 weeks)? LEA is the most common non-pathological cause of HPG suppression.
3. **Audit sleep:** Duration, quality, and consistency. Sleep disruption alone can account for subclinical testosterone suppression.
4. **Evaluate body composition:** Visceral adiposity increases aromatase activity, creating a testosterone-lowering, estrogen-raising cycle.
5. **Check fat intake:** Below 0.5 g/kg/day? This is a modifiable variable with direct impact on steroidogenesis.
6. **Screen micronutrients:** Zinc, magnesium, vitamin D. Replenish if deficient through diet or supplementation.
7. **Optimise training load:** Address any signs of overreaching or undertraining relative to recovery capacity.
8. **Reassess in 8 to 12 weeks:** Repeat labs after lifestyle interventions before considering clinical pathways.

CHAPTER SUMMARY

Testosterone is the primary androgen and a critical anabolic hormone synthesised from cholesterol via the steroidogenic pathway. The HPG axis (GnRH to LH to testosterone) is regulated by negative feedback from testosterone and estradiol. Lifestyle interventions, including energy availability, dietary fat, sleep, body fat management, training load, and micronutrient status, can optimise testosterone within physiological ranges. Free testosterone (not total) is the clinically relevant metric because SHBG determines bioavailability.

◈ MAKE THIS WORK FOR YOU

Testosterone is the most misunderstood hormone in fitness — mostly because of the supplement industry. The honest science: **lifestyle can optimise your testosterone within its natural range**; no over-the-counter product meaningfully raises it above that range. For 95% of men, the "optimisation" protocol is boring, cheap, and effective.

If you are a healthy man with normal levels: The lifestyle levers, in order of evidence: (1) Energy availability — crash dieting tanks T; eat at maintenance or a modest deficit (Chapter 15). (2) Body fat — visceral fat aromatises T to estrogen (Chapter 9); losing belly fat reliably raises T. (3) Sleep — 7–9 hours; one week of sleep restriction measurably drops T (Chapter 13). (4) Dietary fat — keep it above ~0.6–0.9 g/kg/day; very-low-fat diets reduce T. (5) Training — heavy compound lifting has the strongest acute response (Chapter 14); overtraining does the opposite. (6) Micronutrients — zinc, magnesium, vitamin D (test first — Chapter 19). Do all six consistently for 8–12 weeks before drawing any conclusions.

If you suspect low T: The symptoms (libido, energy, mood, strength) overlap heavily with plain lifestyle problems — overtraining, poor sleep, dieting, stress. The correct sequence: fix lifestyle for 8–12 weeks, *then* test (morning, fasted, full panel — total T, free T, LH/FSH, SHBG, estradiol — Chapter 20). If levels are still low with clean lifestyle, see a clinician — but know that the fix for most men is in the lifestyle phase, not a prescription. Do not buy "natural testosterone boosters" — they are marketing built on tiny studies and placebo effects.

If you are an athlete on TRT or considering it: TRT is medical treatment for diagnosed hypogonadism — a legitimate clinical tool, not a bodybuilding shortcut (and its use in sport is regulated). If prescribed: monitor with a clinician (blood work, haematocrit, estradiol), and understand that TRT changes your training response — programme expectations accordingly. Never self-prescribe testosterone; the risks (fertility suppression, cardiovascular, mood) are real and manageable only with medical oversight.

4. Cortisol: The Catabolic-Anabolic Balance

4.1 The Glucocorticoid Axis

Cortisol is the primary glucocorticoid in humans, synthesised in the zona fasciculata of the adrenal cortex. It is regulated by the HPA axis: CRH from the hypothalamus stimulates pituitary ACTH secretion, which activates cortisol synthesis via the cAMP/PKA pathway. Cortisol follows a robust circadian rhythm: peak at 6 to 8 AM (the cortisol awakening response, CAR), declining by 50% by noon, reaching a nadir around midnight. This rhythm is the most reproducible endocrine rhythm in humans and is a sensitive marker of HPA axis health.

4.2 Physiological Roles

Table 4.1 Cortisols Physiological Effects

SYSTEM	EFFECT	RELEVANCE TO COACHING
Carbohydrate metabolism	Gluconeogenesis, glycogenolysis, insulin antagonism	Chronic elevation impairs glucose disposal and glycogen resynthesis
Protein metabolism	Proteolysis (muscle breakdown), reduced MPS	Elevated cortisol directly impairs the anabolic response to training
Fat metabolism	Lipolysis (peripheral) + visceral adipogenesis (chronic)	Chronic elevation shifts fat storage to visceral depots
Immune function	Anti-inflammatory (acute); immunosuppressive (chronic)	Chronic elevation increases infection risk and impairs recovery
Bone metabolism	Inhibits osteoblast activity, stimulates osteoclasts	Long-term elevation compromises bone mineral density
HPG axis	Suppresses GnRH, LH, and gonadal steroidogenesis	Cortisol is the primary endocrine antagonist to the anabolic axis

4.3 The Cortisol-Testosterone Ratio

The cortisol-to-testosterone (C:T) ratio is a more informative metric than either hormone in isolation. A rising C:T ratio reflects a catabolic shift in the hormonal milieu and is associated with: overtraining syndrome, chronic caloric restriction, sleep debt, and psychological burnout. A C:T ratio above 0.3 (measured in compatible units) warrants intervention. Monitoring the C:T ratio over a training mesocycle can identify the early stages of overreaching before performance declines. The recovery of the C:T ratio during a deload week is a marker of adequate recovery capacity.

Practical Application

A client who is stuck despite adequate training and nutrition may have an unfavourable C:T ratio. Before adjusting calories or training variables, assess: sleep quality, life stress, training volume relative to recovery, and psychological load. If the C:T ratio is elevated, the corrective intervention is not more training or more restriction, it is recovery, sleep, stress management, and possibly increasing energy availability.

4.4 HPA Axis Dysregulation

Chronic stress leads to HPA axis dysregulation through two distinct but sequential phases. In **Phase 1 (hypercortisolemia)**, CRH and cortisol are chronically elevated; the feedback system is intact but the set point has shifted upward. In **Phase 2 (HPA hypoactivity)**, the adrenal cortex becomes hyporesponsive to ACTH, and cortisol output drops below normal. The client experiences fatigue, poor stress tolerance, and paradoxical low cortisol on labs despite high perceived stress. This is not adrenal failure, it is central HPA downregulation, and the treatment is reduction of allostatic load, not adrenal supplementation.

4.5 Lifestyle Modulation of Cortisol

Evidence-based interventions to manage cortisol: (1) Sleep extension: even 30 minutes of additional sleep reduces next-day cortisol AUC. (2) Morning light exposure: bright light within 30 minutes of waking entrains the cortisol rhythm and improves the CAR. (3) Training timing: evening high-intensity training can delay the nocturnal cortisol nadir; morning training is preferable for clients with elevated cortisol. (4) Downregulation practices: slow-paced breathing (5 seconds inhale, 5 seconds exhale), meditation, and nature exposure reduce cortisol within 20 minutes. (5) Carbohydrate timing: post-training carbohydrates blunt the cortisol response to training. (6) Caffeine management: caffeine amplifies the cortisol response to stress; limit to under 400 mg/day and avoid after 2 PM.

CHAPTER SUMMARY

Cortisol is the primary glucocorticoid, regulated by the HPA axis with a robust circadian rhythm. It is catabolic to muscle, anabolic to visceral fat, and suppressive to the HPG axis. The cortisol-to-testosterone ratio is a key metric for monitoring overtraining and recovery. HPA dysregulation progresses from hypercortisolemia to central downregulation; treatment requires reducing allostatic load. Six evidence-based interventions can modulate cortisol: sleep, light exposure, training timing, breathing practices, carbohydrate timing, and caffeine management.

◈ MAKE THIS WORK FOR YOU

Cortisol is not the villain — it is the hormone that wakes you and fuels training. The problem is only **chronic elevation**, which catabolises muscle, adds visceral fat, and suppresses the reproductive axis (Chapter 2). The C:T (cortisol:testosterone) ratio is your single best objective recovery monitor — when it climbs persistently, you are overreaching or overstressed, not undertrained.

If you are a hard-training lifter: The six evidence-based levers, in order of practicality: (1) Sleep 7–9 hours (the strongest single lever — Chapter 13). (2) Morning light (10–30 minutes — anchors the healthy cortisol rhythm). (3) Training timing — do not stack hard sessions on top of stress peaks; train when you can recover. (4) Breathing practice (10 minutes of extended-exhale breathwork daily — recovery book Chapter 13). (5) Carbohydrate timing — a carb-containing meal post-session blunts the cortisol response to training. (6) Caffeine management — no caffeine within 8–10 hours of bed, and avoid caffeine before hard sessions if you are already wired (sleep book Chapter 17).

If you are dieting hard: A calorie deficit is a cortisol-raising stressor — this is why deep cuts feel terrible and why cortisol rises during dieting. The protections: keep the deficit moderate (300–500 kcal), take 1–2 week diet breaks every 8–12 weeks (Chapter 15 covers energy availability), keep protein high, and prioritise sleep. If you are in a cut and feel wrecked, the answer is a diet break, not more willpower.

If you suspect HPA dysfunction (chronic fatigue, wired-but-tired): The pattern matters: if morning cortisol is low and evening cortisol is high (inverted rhythm), the axis has been running too hot for too long. The treatment is reducing total load — training volume down 20–30%, stress management, consistent sleep — for 4–8 weeks before any further testing. This is a slow system; it took months to dysregulate and takes months to recover. Do not chase "adrenal fatigue" supplements — the evidence-backed protocol is load reduction, not adaptogen blends (Chapter 19 has the short list that works).

5. Thyroid Hormones and Metabolic Rate

5.1 The Thyroid Axis

The thyroid gland produces two primary hormones: thyroxine (T4, the prohormone, representing approximately 80% of thyroid output) and triiodothyronine (T3, the active hormone, representing approximately 20%). The HPT axis is: TRH (hypothalamus) to TSH (pituitary) to T4/T3 (thyroid). Peripheral deiodinases (D1, D2, D3) convert T4 to the active T3 or the inactive reverse T3 (rT3). This conversion is a critical regulatory point: D2 activity is upregulated by cold exposure and low T4; D3 is upregulated by fasting, illness, and high cortisol, creating a low T3 syndrome that reduces metabolic rate as a protective adaptation.

5.2 Thyroid Hormone Effects on Body Composition

Table 5.1 Thyroid Hormone Effects on Metabolism

TARGET TISSUE	EFFECT	BODY COMPOSITION IMPACT
Skeletal muscle	Increases basal oxygen consumption, mitochondrial biogenesis, myosin ATPase	Low T3 reduces metabolic rate and lipid oxidation
Adipose tissue	Increases lipolysis, UCP1, UCP3 expression	Low T3 reduces fat oxidation, shifts toward fat storage
Liver	Increases gluconeogenesis, LDL receptor expression	Low T3 reduces hepatic clearance, elevates cholesterol
Heart	Increases chronotropic and inotropic activity	Low T3 reduces cardiac output and exercise tolerance

5.3 Low T3 Syndrome (Non-Thyroidal Illness Syndrome)

Low T3 syndrome is characterised by normal TSH, low T3, elevated rT3, and normal/increased T4. It occurs in response to: caloric restriction (reduced D1/D2 activity, increased D3), prolonged exercise, illness, and chronic stress. It is an adaptive response that reduces metabolic rate to conserve energy. In the coaching context, prolonged fat-loss phases (beyond 12 weeks) frequently produce low T3 syndrome, which manifests as a plateau in weight loss despite maintained compliance, cold intolerance, reduced training performance, and mood disturbance. The treatment is not thyroid hormone replacement, it is a structured reverse diet (increasing calories by 50 to 100 kcal/week) or a diet break at maintenance calories for 2 to 4 weeks.

5.4 Thyroid Labs Interpretation

Table 5.2 Thyroid Lab Patterns and Interpretation

TSH	T4	T3	RT3	INTERPRETATION
High	Low	Low/Normal	Normal	Primary hypothyroidism (Hashimotos most common)
Low/Normal	Low	Low	Normal	Central hypothyroidism (pituitary/hypothalamic)
Low	High	High	Low	Hyperthyroidism / thyrotoxicosis
Normal	Normal	Low	High	Low T3 syndrome (NTIS)
High	Normal	Normal	Normal	Subclinical hypothyroidism (TSH above 4.5)

CHAPTER SUMMARY

Thyroid hormones set the basal metabolic rate through T3-mediated effects on muscle, adipose, liver, and heart. Low T3 syndrome is an adaptive response to caloric restriction that reduces metabolic rate and contributes to fat-loss plateaus. Treatment is dietary intervention (reverse diet or diet break), not thyroid hormone replacement. TSH, T4, T3, and rT3 together provide a complete picture of thyroid function.

◈ MAKE THIS WORK FOR YOU

The thyroid sets your metabolic thermostat — and the most important thing to know is that **dieting itself slows it down**. Low T3 during a calorie deficit is a normal adaptive response, not a disease. This explains the classic fat-loss plateau: your metabolism adapts to the deficit, and the fix is a diet break, not thyroid medication.

If you are cutting and stalled: You are probably experiencing metabolic adaptation (low T3 syndrome), not thyroid disease. The evidence-based fix: a diet break — 1–2 weeks at maintenance calories every 8–12 weeks of cutting — restores T3, energy, and adherence, and usually re-ignites fat loss. Or a reverse diet (gradually increasing calories over 4–8 weeks) at the end of the cut. Do not take thyroid medication for diet-induced adaptation — it is not indicated and carries real risks (heart rhythm, muscle loss).

If you have genuine hypothyroid symptoms: Cold intolerance, unexplained weight gain, constipation, dry skin, hair loss, fatigue, slow heart rate — these warrant a full panel: TSH, free T4, free T3, plus antibodies (Hashimoto's is the most common cause) and vitamin D/ferritin (Chapter 20). If diagnosed, medication with medical oversight works excellently — and note that training and protein needs are the same once levels are normalised.

If you are a competitive athlete or female with heavy training: Very low energy availability suppresses T3 (the HPT axis is energy-sensitive — Chapter 15). Chronic under-eating + heavy training + low T3 is a hallmark of RED-S in female athletes (Chapter 22). The fix is energy availability, not thyroid drugs — this is one of the most important coaching distinctions in this book: *dietary treatment, not hormonal treatment.*

The takeaway rule: Thyroid issues fall into two buckets — adaptation (fix with food/diet breaks) and disease (fix with a doctor). If your TSH is normal and you are dieting, your slow metabolism is adaptation. If TSH is out of range, see a clinician. Never self-treat the thyroid; it is one of the few glands where guessing is genuinely dangerous.

6. Insulin, IGF-1 and Growth Hormone

6.1 The Somatotropic Axis

Growth hormone (GH) is secreted by the anterior pituitary in a pulsatile pattern, primarily during slow-wave sleep (the largest pulse) and in response to exercise, fasting, and hypoglycaemia. GH acts directly on tissues (lipolysis, insulin antagonism) and indirectly through hepatic IGF-1 production. IGF-1 mediates the anabolic and growth-promoting effects of GH. The axis is regulated by GHRH (stimulatory) and somatostatin (inhibitory). Ghrelin, secreted by the stomach during fasting, is a potent GH secretagogue.

6.2 GH and IGF-1 in Body Composition

GH promotes lipolysis by activating hormone-sensitive lipase (HSL) in adipose tissue, increasing free fatty acid availability. It is insulin-antagonistic, reducing glucose uptake and promoting hepatic gluconeogenesis. IGF-1 mediates MPS, satellite cell proliferation, and bone growth. The GH/IGF-1 axis is the primary driver of lean mass accretion during a calorie surplus and the primary defender of lean mass during a deficit. Low IGF-1 is associated with reduced MPS, sarcopenia, and impaired recovery from training.

6.3 Insulin Sensitivity and Body Composition

Insulin sensitivity is the ability of cells to respond to insulin and clear glucose. High sensitivity means less insulin is required for a given glucose load, which favours fat oxidation and muscle-directed nutrient partitioning. Strategies to improve insulin sensitivity include: resistance training (GLUT4 upregulation), aerobic training (mitochondrial density), carbohydrate periodisation, fibre intake, reduced visceral adiposity, and adequate sleep.

Key Principle

Insulin is not inherently bad for body composition. The problem is chronically elevated baseline insulin (from frequent feeding, high-glycaemic load without exercise demand, or insulin resistance), not the postprandial spikes that follow carbohydrate-containing meals.

6.4 The GH-IGF-1-Insulin Crosstalk

GH and insulin have antagonistic effects on glucose metabolism: GH promotes lipolysis and insulin resistance, while insulin promotes glucose disposal and lipid storage. This creates a temporal partitioning system: during fasting/exercise/sleep, a GH-dominant state favours fat oxidation; during feeding, an insulin-dominant state favours nutrient storage and MPS. Disruption of this cycle impairs body composition outcomes.

CHAPTER SUMMARY

GH and IGF-1 drive lipolysis and lean mass accretion. Insulin is the master storage hormone directing nutrients into tissues. Insulin sensitivity determines nutrient partitioning. The GH-insulin antagonism creates a natural cycle of lipolysis and storage. Coach: train in the GH-dominant state, feed in the post-training insulin-sensitive window, and maintain overnight fasting to preserve GH pulsatility.

◈ MAKE THIS WORK FOR YOU

Insulin and growth hormone play opposite roles — insulin stores, GH mobilises — and the practical game is **keeping insulin sensitive** so nutrients go to muscle, not fat. Insulin sensitivity is the master variable for body composition; everything that improves it (muscle mass, training, sleep, less visceral fat) compounds.

If you are a lifter: The practical protocols: (1) Train in a GH-favourable state — fasted or low-insulin morning training slightly favours lipolysis during the session (a real but modest effect; train fasted only if you tolerate it — performance trumps the metabolic detail). (2) Feed post-training — the insulin-sensitive window after training is when carbs and protein are partitioned best toward muscle; a post-session meal with protein + carbs (20–50 g carbs is plenty unless you are a high-volume athlete) is the optimal placement. (3) Overnight fasting — 10–14 hours without food preserves the overnight GH pulses and improves next-day insulin sensitivity; this is the least-discussed and most effective "timing" tool (eat dinner 2–3 hours before bed and skip the midnight snacking).

If you are overweight or insulin-resistant (prediabetic pattern): The single most effective intervention is muscle — resistance training is the best insulin-sensitivity drug that exists, because muscle is your main glucose sink. Add: walking after meals (10–20 minutes measurably blunts post-meal glucose), protein at meals, reduced ultra-processed food and sugar-sweetened drinks, and weight loss at a sustainable rate. If you have a family history of diabetes, high waist circumference, or elevated fasting glucose, get a fasting glucose + HbA1c test (Chapter 20) — insulin resistance is the silent condition that explains "I eat less and still gain fat".

If you are lean and cutting: Your insulin sensitivity is probably fine — do not chase low-carb extremes. The rule for lean people: carbs are your friend around training; keep them there (pre/post-session) and they will not impair fat loss. Overnight fasting and the post-training feeding window are the two timing tools that matter; the rest is noise.

Coaching Application

Domain 1 Case Study: The Stuck Fat Loss Client

Scenario: A 29-year-old female client, 68 kg, 26% body fat, has been on a calorie deficit (1,800 kcal/day) for 14 weeks. She trains 4x/week (hypertrophy and LISS). Initial weight loss of 5 kg in the first 6 weeks has stalled completely for the past 5 weeks. She reports cold extremities, low energy, poor sleep, and reduced training performance. Diet adherence is confirmed.

Assessment: Classic low T3 syndrome from prolonged caloric restriction. Expected labs: normal TSH, low T3, elevated rT3, elevated cortisol, low-normal free testosterone, low leptin. The HPA/HPT axes have downregulated metabolic rate in response to chronic energy deficit.

Intervention: (1) 4-week diet break at estimated maintenance (2,200 kcal/day) to normalise T3, leptin, and cortisol. (2) Maintain protein at 2.0 g/kg (136 g/day). (3) Reduce training intensity (RPE 6 to 7) to reduce additional cortisol stimulus. (4) Increase sleep to 8+ hours. (5) After 4 weeks, resume deficit at a smaller magnitude (200 to 300 kcal/day deficit). (6) Add a structured refeed day per week during the new deficit phase.

Key Takeaway: Prolonged deficit induces metabolic and hormonal adaptation that resists further weight loss. The corrective intervention is not further restriction, it is a diet break to normalise the hormonal milieu, followed by a smaller, more sustainable deficit with strategic refeeds.

DOMAIN 2

Sex Hormones and Reproductive Endocrinology: estrogen, progesterone, SHBG, androgens, and the complete gonadotropin axis

7. Estrogen, Progesterone and the Menstrual Cycle

7.1 The Ovarian Steroidogenic Axis

Estrogen and progesterone are the primary female sex hormones, synthesised in the ovarian follicles and corpus luteum, respectively. The HPG axis regulates their production: GnRH to FSH (follicular development) to estrogen; GnRH to LH (ovulation, luteinisation) to progesterone. Estrogen exists in three forms: estradiol (E2, the primary and most potent), estrone (E1, dominant post-menopause), and estriol (E3, primarily during pregnancy). E2 is

Follicular (late)	6 to 13	Rising E2	Endometrial thickening; E2 improves insulin sensitivity and mood
Ovulation	14	E2 peak to LH surge	LH surge triggers oocyte release; E2 peak correlates with peak strength
Luteal (early)	15 to 21	Rising P4, moderate E2	Corpus luteum secretes progesterone; P4 is catabolic and thermogenic
Luteal (late)	22 to 28	P4 peak to drop, E2 drop	P4 decline triggers menstruation; reduced exercise tolerance

7.3 Estrogens Anabolic Role

Estrogen is not merely a female hormone, it has significant anabolic and metabolic effects. E2 receptors (ERa, ERb) are expressed in skeletal muscle, bone, and adipose tissue. E2 promotes: (1) satellite cell activation and muscle repair through ERb-mediated IGF-1 signalling, (2) bone mineral density by inhibiting osteoclast activity, (3) subcutaneous fat deposition (gynoid pattern), (4) insulin sensitivity via GLUT4 upregulation, and (5) GH pulsatility. The drop in E2 at menopause is associated with accelerated muscle loss, bone loss, and a shift to visceral adiposity.

7.4 Progesterone: The Modulator

Progesterone tempers many of estrogens effects. It is thermogenic (raising body temperature by 0.3 to 0.5 degrees C in the luteal phase), mildly catabolic, and promotes fluid retention. In the luteal phase, many women experience reduced exercise tolerance, impaired recovery, and increased perceived exertion. This is not pathological, it is the normal hormonal signature of the luteal phase. The coaching response is to adjust expectations and programming, not to pathologise the cycle.

Coaching Takeaway

The menstrual cycle produces predictable shifts in performance, recovery, and mood. The follicular phase is generally associated with better performance and recovery. The luteal phase is associated with reduced exercise tolerance and higher RPE at submaximal loads. Programming should acknowledge these shifts without rigid over-prescription. The primary coaching goal is to help the client understand and work with their cycle, not fight it.

CHAPTER SUMMARY

Estrogen and progesterone regulate the menstrual cycle through a predictable hormonal timeline. Estrogen is anabolic, promoting muscle repair, bone density, insulin sensitivity, and GH pulsatility. Progesterone is catabolic and thermogenic, reducing exercise tolerance in the luteal phase. Coaching: monitor individual response, adjust training load in the luteal phase if needed, and use the follicular phase for higher-intensity work.

◈ MAKE THIS WORK FOR YOU

The menstrual cycle is a 28-day hormonal rhythm (21–35 is normal) with two phases: **follicular** (days 1–14 — estrogen rising; generally the "feel good, train hard" phase) and **luteal** (days 15–28 — progesterone dominant; often the "heavier, hotter, slower" phase). The coaching truth: effects are real on average but **highly individual** — some women notice big differences, many notice none. Track before you adjust.

If you are a female athlete who trains hard: The evidence-based approach: (1) Track your cycle + training performance for 2–3 full cycles before changing anything (period-tracking apps work; a simple log is fine). (2) If you notice a pattern (e.g., heavy lifts feel terrible in the late luteal phase), programme accordingly: higher intensity/volume in the follicular phase, technique/moderate work in the late luteal phase. (3) If you notice no pattern, train consistently — forcing cycle-synced programming on a non-responder is wasted effort. (4) Practical luteal-phase adjustments that help most women who do experience dips: longer warm-ups, slightly lower training expectations (RPE runs higher), more sleep and carbs (luteal phase increases carbohydrate needs slightly), and more patience — the same weights will feel light again in the next follicular phase.

If you coach female clients: Do not presume — ask. A client's cycle is a legitimate programming variable only if she experiences symptoms. And never use "it's your cycle" as a dismissal: if a client reports a pattern change (periods heavier/painful/absent, training suddenly harder), that is a signal for medical review, not reprogramming (Chapters 11 and 20).

If your cycle is irregular or absent: Missing 3+ consecutive periods (or >35-day cycles) outside of pregnancy/contraception = medical flag (RED-S screening — Chapter 22). This is the single most important coaching alert in this book: a lost cycle is the female body's emergency brake, and training through it worsens the damage (bone health in particular). See a clinician; in the meantime, raise energy availability and reduce training load.

8. Free vs. Bound Hormones: SHBG and Bioavailability

8.1 The Binding Protein System

Steroid hormones circulate in three fractions: (1) free (unbound, approximately 1 to 4% of total), (2) albumin-bound (approximately 30 to 40%), and (3) SHBG-bound (approximately 50 to 60%). The free fraction is the primary bioactive pool. The albumin-bound fraction dissociates rapidly in capillary beds and is also bioavailable. Therefore, free plus albumin-bound equals bioavailable hormone, which is the most clinically relevant metric. SHBG binds testosterone and estradiol with high affinity and slow dissociation, rendering the SHBG-bound fraction effectively inactive.

8.2 SHBG Regulation

Table 8.1 Factors That Modulate SHBG

FACTOR	EFFECT ON SHBG	MECHANISM
Estrogen	Increases SHBG	E2 upregulates hepatic SHBG synthesis
Testosterone	Decreases SHBG	Androgens suppress hepatic SHBG production
Insulin	Decreases SHBG	Insulin suppresses HNF-4a, the SHBG transcription factor
Thyroid hormones (T3)	Increase SHBG	T3 upregulates SHBG gene expression
Obesity / Insulin resistance	Decreases SHBG	Hyperinsulinaemia suppresses SHBG
Caloric restriction	Increases SHBG	Reduced insulin, increased GH
Liver disease	Decreases SHBG	Reduced hepatic synthetic capacity

8.3 Why Total Testosterone Can Be Misleading

A male client may have total testosterone of 600 ng/dL (well within range) but very low free testosterone due to elevated SHBG. This scenario is common in chronic dieters, endurance athletes, and individuals with hyperthyroidism or estrogen dominance. The total T looks normal, but the client experiences symptoms of low testosterone. Conversely, a client with low total T but low SHBG (common in obesity/insulin resistance) may have normal free T and fewer symptoms. This is why free testosterone must be measured alongside total T to assess the true hormonal state.

8.4 Calculated Free Testosterone

The equilibrium dialysis method is the gold standard for free T measurement, but the Vermeulen equation is a reliable clinical surrogate. The formula requires total T and SHBG drawn simultaneously, and online calculators are widely available. The key coaching point: free T calculation requires total T and SHBG to be drawn together.

CHAPTER SUMMARY

Steroid hormones circulate as free, albumin-bound, and SHBG-bound fractions. Only free and albumin-bound hormone is bioavailable. SHBG is modulated by estrogen (up), androgens (down), insulin (down), and T3 (up). Total testosterone can be misleading. Always assess free testosterone alongside total T to understand the true hormonal picture.

◈ MAKE THIS WORK FOR YOU

Here is the subtlety that explains most "my testosterone is fine but I feel terrible" mysteries: **total hormone levels are not the whole story** — only the free (bioavailable) fraction acts on your tissues. A normal total T with high SHBG (the binding protein) means low free T — the "bound up" pattern.

If you are planning blood work: Always ask for total AND free testosterone (plus SHBG so you can see the relationship), not just total. In women, the same applies to estradiol and testosterone interpretation. The classic scenarios: total T normal + SHBG high = functionally low free T (symptoms without "low" labs) — often seen with low-carb dieting (insulin down → SHBG up), oral contraceptives (estrogen up → SHBG up — Chapter 23), and hyperthyroid states. Total T low + SHBG low = different problem (production), Chapter 20 interprets the combinations.

If you are a male lifter dieting low-carb: Prolonged very-low-carb dieting can raise SHBG and lower free T even with normal total T. The practical fix if you are low-energy on a low-carb cut: reintroduce some carbohydrates (especially around training) and re-test after 4–6 weeks — this is a diet issue, not a testosterone disease. Insulin sensitivity and SHBG respond within weeks.

If you are a female on the combined pill: COCs raise SHBG 2–3x, which binds more free testosterone — this is why oral contraceptives can lower libido and blunt the anabolic environment (Chapter 23 covers the full athlete picture). It is a medication trade-off made with your doctor, but knowing the mechanism lets you interpret symptoms and have an informed conversation rather than accepting "it's just the pill".

The takeaway: When interpreting hormones, think in terms of *free, bioavailable* hormone and the SHBG context — a "normal" total number with symptoms is worth a second look at the free fraction and binding proteins before you conclude anything.

9. Aromatase, DHT and Androgen Metabolism

9.1 The Androgen Metabolic Pathways

Testosterone is metabolised through two primary pathways: (1) aromatisation to estradiol (E2) via the aromatase enzyme (CYP19A1), and (2) 5a-reduction to dihydrotestosterone (DHT) via 5a-reductase. These two pathways determine the hormonal signature of testosterone activity in different tissues. Aromatase is expressed in adipose tissue, bone, brain, breast, and testis. 5a-reductase is expressed in the prostate, skin, hair follicles, and liver. The balance between these pathways determines the clinical phenotype.

9.2 DHT: The Potent Androgen

DHT is approximately 3 to 5 times more potent than testosterone as an androgen due to higher AR binding affinity and slower dissociation. DHT cannot be aromatised, making it a pure androgen. Its effects include: prostate growth, male pattern hair growth/loss, sebaceous gland activity, and libido. DHT is not anabolic for skeletal muscle in physiological doses, testosterone itself is the primary muscle androgen. 5a-reductase inhibitors (finasteride, dutasteride) reduce DHT and can improve scalp hair retention but may cause side effects including reduced libido and erectile dysfunction.

9.3 Aromatase Activity and the E2:T Ratio

Table 9.1 Factors Affecting Aromatase Activity

FACTOR	EFFECT ON AROMATASE	EFFECT ON E2:T RATIO
Visceral adiposity	Upregulates CYP19A1 expression	Increases (more T to E2)
High body fat (%)	More adipose tissue equals more aromatisation	Increases
Alcohol consumption	Increases hepatic aromatase activity	Increases
Zinc deficiency	Indirect: zinc inhibits aromatase	Increases
Zinc supplementation	Inhibits aromatase activity	Decreases
Flavonoids / grape seed extract	Mild aromatase inhibition	Decreases (modest)

9.4 Androgen Receptor Sensitivity

AR sensitivity, determined by CAG repeat length in the AR gene, is a significant determinant of individual response to androgens. Shorter CAG repeats correlate with higher AR sensitivity and greater myotrophic response to testosterone. This genetic polymorphism explains a portion of individual variability in training response. It is clinically relevant but not routinely tested in coaching practice.

CHAPTER SUMMARY

Testosterone is metabolised via aromatisation (to E2) and 5a-reduction (to DHT). Visceral adiposity increases aromatase activity, raising the E2:T ratio and contributing to testosterone suppression. DHT is a pure androgen with minimal direct anabolic effect on muscle. Body fat management is the most effective modifiable intervention for optimising the E2:T ratio.

◈ MAKE THIS WORK FOR YOU

Testosterone converts into two other hormones: **estradiol (E2)** via aromatase — which is fine in the right balance (men need some E2 for bones, brain, and libido) — and **DHT**, the potent androgen behind hair, skin, and prostate. The balance is heavily influenced by body fat: fat tissue is where aromatase lives, so more visceral fat = more conversion of T to E2 = a feedback loop that suppresses T further.

If you are overweight (the most common pattern): This is the mechanism behind the "fat man low-T" loop: visceral fat → more aromatase → more E2 → HPG suppression → lower T → harder to lose fat. The good news: the loop reverses. Weight loss (even 5–10% of body weight) reliably lowers E2, raises T, and improves the ratio. This is the single most effective "hormonal optimisation" available to overweight men — and it requires no supplements. Waist circumference is your tracking metric (target < 94 cm / < 80 cm for men/women).

If you are lean with hormonal symptoms: A lean male with low T or high E2 symptoms (breast tenderness, low libido, ED) should get the full picture: total/free T, E2 (sensitive assay), SHBG, LH/FSH, prolactin — and check lifestyle first (sleep, training volume, alcohol — heavy drinking raises aromatase and E2). If lifestyle is clean and labs are odd, see a clinician — aromatase inhibition is a medical (not supplement) domain; "aromatase inhibitor" supplements are unproven and can be harmful. DHT, for the record, is not something you need to "boost" — it is an effect hormone; hairloss concerns with DHT-related genetics are a dermatology conversation, not a fitness one.

If you are a female athlete: Androgen metabolism matters in reverse: excess androgen activity (PCOS, adrenal) presents as acne, hirsutism, and irregular cycles — a medical diagnosis, not a fitness problem. Do not self-treat with supplements; PCOS responds well to the same core tools (weight management, resistance training, insulin sensitivity — Chapter 6) but needs medical confirmation.

10. Prolactin, LH and FSH: The Gonadotropin Axis

10.1 The Gonadotropins

LH and FSH are glycoprotein hormones secreted by the anterior pituitary in response to GnRH. In males, LH stimulates Leydig cells to produce testosterone; FSH supports spermatogenesis. In females, LH triggers ovulation; FSH stimulates follicular growth. Measuring LH and FSH differentiates primary hypogonadism (high LH/FSH, indicating the pituitary is responding to low peripheral hormone) from secondary hypogonadism (low/inappropriately normal LH/FSH, indicating central suppression).

10.2 Prolactin: The Modulator

Prolactin is unique among pituitary hormones in that its release is tonically inhibited by dopamine (PIF). When dopamine signalling is disrupted by stress, certain medications, pituitary tumours, or hypothyroidism (TRH stimulates prolactin), prolactin rises and suppresses GnRH, LH, and FSH, causing hypogonadism. Elevated prolactin is a treatable cause of low libido and hypogonadism in both sexes.

10.3 Common Lab Patterns

Table 10.1 Gonadotropin Patterns and Interpretation

TOTAL T	LH	FSH	PROLACTIN	INTERPRETATION
Low	Low/Normal	Low/Normal	Normal	Secondary hypogonadism (functional: stress, LEA, sleep, obesity)
Low	High	High	Normal	Primary hypogonadism (testicular failure; rare in young athletes)
Low	Low/Normal	Low/Normal	High	Hyperprolactinaemia; check pituitary MRI if prolactin above 50
Normal/Low	Very Low	Low	Normal	Hypogonadotropic hypogonadism; suspect opioid use or GnRH deficiency

CHAPTER SUMMARY

LH and FSH differentiate primary from secondary hypogonadism. Low LH/FSH with low T indicates central (functional) suppression, the most common pattern in athletes and dieters. Prolactin is tonically inhibited by dopamine; elevated prolactin suppresses the HPG axis and is a treatable cause of low libido and hypogonadism.

◈ **MAKE THIS WORK FOR YOU**

LH and FSH are the pituitary's "production orders" to the gonads — and measuring them tells you **where** a hormone problem lives: if LH/FSH are low alongside low T, the problem is upstream (brain/lifestyle — the common athlete pattern); if they are high alongside low T, the problem is the gland itself (medical). This distinction decides whether your fix is lifestyle or a clinic.

If you are an athlete or dieter with low T: The classic finding is central (functional) suppression — low LH/FSH + low T — caused by: chronic energy deficit (the #1 athlete cause — Chapter 15), overtraining, sleep deprivation, or chronic stress. The good news: this pattern is *reversible*. Fix energy availability, cut training volume during high-stress periods, and sleep 7–9 hours — retest in 8–12 weeks and watch the axis recover. This is the most common "low T" in the gym population, and it is rarely a medical disease.

If your libido has dropped: Prolactin matters more than most people know — it is the one hormone that directly suppresses the HPG axis, and elevated prolactin is a *treatable* cause of low libido and erectile dysfunction in men (and cycle disruption in women). It is raised by: stress, sleep deprivation, some medications (antidepressants, antipsychotics), and rarely pituitary adenomas (with nipple discharge — an urgent flag). If you test and prolactin is high, this is a doctor conversation — it is one of the most fixable endocrine problems that exists, so do not sit on it.

If you are planning testing: Include LH/FSH and prolactin in any comprehensive male panel (Chapter 20's minimal panel has them) — the interpretation power they add is worth the small extra cost, because "low T" without LH/FSH is like a car warning light without the manual.

11. Female Hormonal Health Across the Lifespan

11.1 The Female Hormonal Trajectory

Female hormonal health evolves through distinct life stages: menarche, regular ovulatory cycles, perimenopause (age 35 to 50), menopause (approximately 51 on average), and post-menopause. Each stage presents unique considerations for body composition coaching, and the hormonal profile at each stage influences the expected response to training and nutrition interventions.

11.2 Low Energy Availability and the Female Athlete

The Female Athlete Triad has been expanded to Relative Energy Deficiency in Sport (RED-S), which encompasses a broader range of consequences of low energy availability (LEA). In females, LEA suppresses the HPG axis at higher energy thresholds than males: menstrual disruption can occur below 30 to 45 kcal/kg FFM/day, compared to the male threshold of below 20 to 25. Consequences include: menstrual dysfunction (functional hypothalamic amenorrhea), reduced bone mineral density, impaired MPS, increased injury risk, and immune dysfunction.

11.3 Perimenopause and Menopause in Body Composition

The perimenopausal period is a window of increasing coaching relevance. Key principles: (1) Prioritise resistance training as the primary intervention for muscle and bone preservation. (2) Protein intake must be higher (1.8 to 2.4 g/kg/day) to overcome anabolic resistance from E2 decline. (3) Calorie restriction must be approached cautiously, the hormonal milieu is already catabolic. (4) Sleep disruption is a primary mediator; addressing sleep quality is the prerequisite. (5) HRT is a medical decision but, when appropriate, can significantly improve the response to lifestyle interventions.

CHAPTER SUMMARY

Female hormonal health evolves from menarche through menopause. LEA suppresses the female HPG axis at higher thresholds than males. Perimenopause and menopause present challenges: E2 decline reduces the anabolic ceiling and increases visceral fat tendency. Coaching: resistance training, higher protein, sleep prioritisation, and cautious energy balance.

◈ MAKE THIS WORK FOR YOU

Female hormonal life is a sequence of phases — menarche, the reproductive years, pregnancy, perimenopause, menopause — and the coaching rules shift at each transition. The most important athlete-specific fact in this chapter: **the female HPG axis is more energy-sensitive than the male's** — low energy availability (LEA) suppresses the cycle at thresholds men never experience. This is the root of RED-S (Chapter 22).

If you are a female athlete in your reproductive years: Your cycle is a health monitor, not a nuisance: a regular cycle means energy availability is adequate and the axis is healthy. Track it (apps or a simple log). If it becomes irregular or stops, that is your body's alarm — energy availability, training load, and stress are the usual suspects (Chapter 22's RED-S screen). Protect the fundamentals: adequate calories around training, fat intake not too low, iron status checked annually (especially with heavy periods — ferritin below ~30 ng/mL is common and saps training performance), and sleep.

If you are perimenopausal (usually 40s–50s): E2 fluctuates and then declines — symptoms include disrupted sleep (night sweats), weight gain tendency (especially visceral), mood changes, and reduced training capacity at the top end. This is a coaching-relevant transition: resistance training becomes *more* important (bone density, muscle preservation, metabolic rate), protein should rise (1.6–2.0 g/kg), and sleep work (Chapter 9 temperature protocols of the sleep book) is essential. HRT is safe and effective for many women and dramatically underused — an informed conversation with a menopause-informed clinician is strongly worth having; symptoms are treatable, not "just aging".

If you are post-menopausal: The anabolic ceiling is lower and visceral fat tendency higher — but the tools are identical and highly effective: resistance training 2–3x/week (non-negotiable — bone and muscle), protein top of range, vitamin D + calcium adequacy (bone health), and weight management with a bias toward resistance over endless cardio. Women who train through menopause age far better than those who don't — this phase rewards consistency more than any supplement.

12. Libido, Fertility and Hormonal Health Markers

12.1 Libido as a Coaching Metric

Libido is a sensitive indicator of hormonal health that often declines before measurable changes in serum hormone concentrations. It reflects the integrated status of the HPG axis, dopamine tone, prolactin, cortisol, and psychological state. A sudden or sustained decline in libido in the absence of relationship factors should prompt a hormonal investigation. In males, libido correlates more strongly with free testosterone than total testosterone.

12.2 Fertility Markers

LH and FSH should be measured alongside testosterone to assess the integrity of the HPG axis. Elevated FSH (above 10 IU/L) indicates impaired spermatogenesis. In females, regular ovulatory cycles are a proxy for balanced hormonal health. Cycle tracking is a coaching tool that can identify hormonal dysfunction before it manifests clinically.

12.3 Hormonal Health Markers Checklist

Table 12.1 Quick Hormonal Health Screening Questions

QUESTION	ASSESSES	FOLLOW-UP
Are you sleeping 7 to 9 hours/night?	HPA axis, GH, HPG axis	If no: sleep protocol required first
Do you wake refreshed or fatigued?	Cortisol rhythm	Poor CAR correlates with HPA dysregulation
How is your libido vs. 6 months ago?	HPG axis	Decline warrants hormonal investigation
Are your cycles regular? (female)	HPG axis integrity	Irregular above 35-day variability indicates disruption
Do you feel recovered from training?	Recovery capacity, C:T ratio	Persistent non-recovery suggests hormonal dysfunction
Are you losing weight as expected?	Thyroid axis	Unexplained plateau suggests low T3 syndrome
Do you feel stressed most days?	HPA axis, allostatic load	Requires stress management intervention

Coaching Perspective

Libido and cycle regularity are free, real-time hormonal biomarkers that precede lab abnormalities. A sudden libido drop in a male client whose total testosterone is 500 ng/dL tells you more than the lab value. These subjective markers should trigger investigation before symptoms progress to measurable lab abnormalities.

CHAPTER SUMMARY

Libido is a sensitive, real-time hormonal health marker. Fertility markers provide deeper insight into HPG axis integrity. The hormonal health screening checklist provides a quick assessment of major endocrine axes. Subjective markers precede lab abnormalities and should trigger proactive investigation.

◈ MAKE THIS WORK FOR YOU

Libido is the body's real-time hormonal dashboard — it drops before labs change and recovers before labs normalise. Used properly, it is a free early-warning system: **a sustained drop in libido (4+ weeks) is one of the first signs of a hormonal or recovery problem** — and often the first visible sign of overtraining, under-eating, or sleep debt.

The screening checklist (run this when libido/energy/mood are off): (1) Training: 4+ hard weeks without a deload? (2) Nutrition: dieting hard or very low calories for 6+ weeks? Fat intake very low? (3) Sleep: under 7 hours consistently, or poor quality? (4) Stress: work/life load high? (5) Alcohol: regular drinking? (6) Weight: rapid loss or gain recently? (7) Medications: antidepressants (libido is a well-known side effect — never stop without medical advice)? If any are yes, fix them first — most "mystery libido drops" resolve with the lifestyle fixes. If everything is clean and libido stays flat for 4+ weeks, run blood work (Chapter 20's panel — including prolactin, Chapter 10).

If you are trying to conceive (male or female): Fertility is the deepest HPG-axis integrity test — and lifestyle is the treatment: men — avoid heat (saunas/hot baths are temporary sperm suppressors), limit alcohol, manage stress, sleep well, and consider zinc/selenium adequacy; both sexes — normalise body weight, avoid crash dieting (fertility is energy-sensitive), and reduce training volume if it is extreme (overreaching impairs sperm quality and ovulation). If 12 months of trying (6 if over 35) without success: see a fertility specialist — this is medical territory, and coaching can support but not substitute for it.

The key principle: Subjective markers (libido, energy, mood, cycle) change *before* blood markers — they are your first-line monitoring, labs are your confirmation. A client or athlete who reports a sustained libido drop is not being embarrassing — they are reporting the most honest hormonal data available. Respond to the signal.

Coaching Application

Domain 2 Case Study: The Hypogonadal Male Client

Scenario: A 34-year-old male, 88 kg, 18% body fat, trains 5x/week (PPL split). He reports low libido for 6 months, poor sleep, irritability, and plateaued gym progress. He has been in a calorie deficit (2,200 kcal/day) for 16 weeks. Labs: total T 320 ng/dL, free T 45 pg/mL, SHBG 45 nmol/L, LH 3.2 IU/L, prolactin 18 ng/mL, cortisol (AM) 28 mcg/dL.

Assessment: Secondary (functional) hypogonadism driven by prolonged caloric deficit and high allostatic load. Low-normal LH suggests inadequate GnRH drive. The 16-week deficit has induced metabolic adaptation that suppressed the entire HPG axis.

Intervention: (1) End the deficit phase, increase to maintenance (2,800 to 3,000 kcal/day) for 8+ weeks. (2) Protein 2.2 g/kg, fat 0.9 g/kg. (3) Reduce training volume by 20%, RPE 7 max. (4) Sleep protocol: 8+ hours. (5) Supplement: zinc 30 mg, magnesium glycinate 400 mg, vitamin D 3,000 IU. (6) Recheck labs at 8 weeks.

Key Takeaway: Prolonged caloric deficit causes functional hypogonadism through LEA-mediated HPG suppression. The treatment is ending the deficit, normalising energy availability, and reducing allostatic load.

DOMAIN 3

Lifestyle and Environmental Modulation: sleep, training, nutrition, stress, circadian biology, and targeted supplementation

13. Sleep Architecture and Hormonal Pulsatility

13.1 Sleep Architecture

Sleep cycles through NREM (N1, N2, N3/slow-wave sleep) and REM stages, with 4 to 6 cycles per night (approximately 90 minutes each). Slow-wave sleep (SWS) is the most hormonally active stage: GH secretion peaks during SWS (approximately 70% of daily GH), cortisol is at its nocturnal nadir, and parasympathetic tone is maximal. REM sleep is associated with HPA axis activation and programming of the cortisol awakening response.

Prolactin	Peaks 4 to 6 AM	SWS-dependent	Blunted prolactin leads to impaired immune recovery
LH (male)	Pulsatile through night	REM-linked pulses	Reduced LH pulse amplitude leads to reduced morning T
Testosterone	Peak upon waking (6 to 8 AM)	Reflects overnight LH	Minus 10 to 15% after less than 5h; minus 30% after severe restriction
Leptin	Peaks approximately 3 to 4 AM	Feeding-independent	Reduced leptin leads to increased hunger, reduced satiety
Ghrelin	Suppressed during sleep	NREM	Increased ghrelin leads to increased appetite, reduced SWS

13.3 The Sleep-Hormone Cascade

Sleep deprivation (less than 7 hours/night) initiates a cascade: (1) Reduced SWS leads to blunted GH pulse and reduced lipolysis and MPS. (2) Extended wakefulness leads to elevated nocturnal cortisol and insulin resistance. (3) Elevated cortisol leads to HPG suppression and reduced morning testosterone. (4) Reduced leptin plus elevated ghrelin leads to increased hunger. (5) Impaired glycogen resynthesis leads to reduced training performance. A single night of poor sleep measurably impairs next-day hormonal function.

13.4 Sleep Optimization Protocol

Sleep Protocol for Hormonal Optimization

1. **Duration:** 7 to 9 hours/night. Consistency within 30-minute bedtime window.
2. **Darkness:** Complete darkness for melatonin. Blackout curtains, eye mask, no blue light 60 to 90 min before bed.
3. **Temperature:** 18 to 21 degrees C promotes SWS. Core temperature drop signals sleep onset.
4. **Eating window:** No large meals within 2 to 3 hours of bed. Pre-sleep casein (30 to 50 g) is beneficial. Avoid alcohol.
5. **Caffeine cutoff:** No caffeine after 2 PM (half-life 4 to 6 hours).
6. **Wind-down routine:** 30 to 60 minutes of low-stimulation activity before bed.
7. **Supplements (conditional):** Magnesium glycinate 200 to 400 mg. Glycine 3 to 5 g. Melatonin 0.5 to 3 mg for shift work only.

CHAPTER SUMMARY

Sleep cycles through NREM/REM stages with distinct hormonal signatures. Sleep deprivation triggers a cascade of hormonal disruption. The sleep protocol emphasises duration, consistency, darkness, temperature, eating window, and wind-down routine. Sleep is the single most impactful non-pharmacological intervention for hormonal optimisation.

◈ MAKE THIS WORK FOR YOU

This is the chapter where the whole system connects: **sleep is the master hormonal switch**. One week of sleep restriction measurably drops testosterone (by ~10–15% in controlled studies), raises evening cortisol, blunts GH pulses, worsens insulin sensitivity, and raises hunger hormones. No supplement, protocol, or training hack compensates for chronic sleep debt — which is why this book's sleep pillar is a prerequisite for the hormonal pillar.

If you are a male lifter: The testosterone-sleep link is direct and fast: a single night of poor sleep has measurable effects; a week of restriction is clearly visible in labs. The practical standard: 7–9 hours, consistent timing (a consistent wake time is the strongest anchor — sleep book Chapter 1), and alcohol discipline (alcohol before bed suppresses both deep sleep and the HGH/testosterone cascade). If you are training hard and feel flat, audit sleep *before* testing hormones — a sleep fix is cheaper and often sufficient (Chapter 13 of the sleep book has the N-of-1 protocol).

If you are a female athlete: Sleep disruption interacts with the menstrual cycle (luteal-phase sleep is often worse — sleep book Chapter 25), and poor sleep worsens cycle regularity under training stress. The practical priorities: consistent sleep timing, temperature management (night sweats need the Chapter 9 toolkit), and treating sleep like a training session in your schedule — especially in luteal weeks.

If you are a coach: The coaching order is fixed: sleep → energy availability → stress → training → supplements → labs. When a client's "hormonal" symptoms appear (low libido, fatigue, poor recovery, stalled progress), the first interventions are always sleep and energy intake — the lab is confirmation, not the starting point. This ordering saves clients thousands in unnecessary testing and supplements.

14. Training Hormonal Signature

14.1 Acute Hormonal Response to Training

Resistance training produces an acute hormonal surge: catecholamines rise, followed by GH, cortisol, and testosterone. These acute elevations are permissive signals that create a transient anabolic window. The magnitude of the acute response is not a measure of session quality, chronic adaptation depends on the cumulative hormonal milieu over days and weeks.

14.2 Training Variables and Hormonal Response

Table 14.1 Training Variables and Hormonal Response

VARIABLE	EFFECT ON ACUTE HORMONES	COACHING IMPLICATION
High volume (10+ sets/muscle)	Elevated GH, cortisol; moderate T	Volume must be periodised; sustained high volume elevates cortisol
High intensity (above 85% 1RM)	Elevated catecholamines, T; low GH	Neural-focused sessions; less metabolic signal
Short rest (less than 60 s)	Elevated GH, cortisol, lactate	Metabolic stress increases cortisol; use selectively
Large muscle mass exercises	Greater total hormonal response	Compounds produce the largest acute surge
Duration above 75 minutes	Cortisol rises disproportionate	Diminishing returns beyond 75 minutes
Training to failure	Elevated cortisol, catecholamines	Catabolic signal disproportionate to anabolic

14.3 Chronic Training Adaptation

Chronic resistance training improves androgen receptor content and signalling efficiency rather than raising resting testosterone. Overtraining syndrome is characterised by declining resting testosterone, elevated resting cortisol, and blunted acute hormonal response to a standardised exercise bout, the C:T ratio shifts unfavourably. Endurance training in high volume suppresses resting testosterone through HPG axis suppression and increased clearance.

14.4 Designing Training for Hormonal Health

Session duration is recommended at or below 60 to 75 minutes. Volume should be periodised with planned deloads every 4 to 6 weeks. Most sets should fall at RPE 7 to 9 (1 to 3 RIR). Compound movements should be prioritised in the first half of the session; isolation work and finishers belong in the second half. Rest intervals should be adequate: 2 to 3 minutes for compounds, 60 to 90 seconds for accessories. A consistent training schedule helps entrain circadian rhythms.

CHAPTER SUMMARY

Resistance training produces an acute hormonal surge that is permissive, not causative, of adaptation. Overtraining is marked by a suppressed C:T ratio. Training for hormonal health: sessions of 75 minutes or less, periodised volume, RPE 7 to 9, compound lifts prioritised, adequate rest intervals, and consistent scheduling.

◈ MAKE THIS WORK FOR YOU

The honest science: the post-lift hormone spike (testosterone, GH, cortisol) does not directly build muscle — it is a *permissive* signal; what builds muscle is the training stimulus itself plus recovery. You do not need "hormone-spiking" workouts. What you DO need is training that does not chronically suppress the hormonal environment — and that is a recovery question, not an intensity question.

If you are a hard-training lifter: The evidence-based training prescription for hormonal health: sessions ≤ 75 minutes (longer sessions add cortisol without adding stimulus), compound lifts prioritised (they generate the largest acute response and best efficiency), RPE 7–9 for working sets (failure-every-set adds fatigue without adding gain), rest intervals 2–3 minutes on big lifts, and periodised volume (not endless accumulation). The C:T ratio is your monitor: if you feel flattened, sleep poorly, and strength stalls across all lifts for 3+ weeks, you are overreaching — deload (recovery book Chapter 23), and the ratio recovers.

If you are overtrained or chronically under-recovered: The single most important sentence in this chapter: *more training is never the fix for suppressed hormones — less is.* Overtraining suppresses the HPG axis (low T, low libido, flat mood). The protocol: reduce volume 20–30%, keep a couple of heavy compound sets at RPE 7–8 (maintain the stimulus without the load), fix sleep and food, and expect 2–6 weeks for the hormonal environment to normalise. Overtrained athletes recover with rest, not with adaptogens (recovery book Chapter 18).

If you are a beginner: None of this should slow you down — your untrained adaptation is driven by progressive overload and consistency, not hormonal fine-tuning. Train 2–4x/week, progress sensibly, sleep well, and your hormones will take care of themselves. The hormonal optimisation layers are for later, when margins are thinner.

15. Nutrition for Hormonal Health

15.1 Energy Availability as the Primary Variable

Energy availability (EA) = (Energy Intake minus Exercise Expenditure) / kg FFM. Low EA (below 30 kcal/kg FFM/day in females, below 20 to 25 in males) suppresses the HPG axis, reduces T3, elevates cortisol, and suppresses leptin. Aggressive or prolonged calorie restriction almost invariably induces hormonal dysfunction.

Table 15.1 Energy Availability Thresholds

EA LEVEL (KCAL/KG FFM/DAY)	HORMONAL EFFECTS	CONTEXT
Above 45	Optimal; full axis function	Maintenance or surplus
30 to 45	Mild HPG suppression; T3 minus 10 to 20%; cortisol plus 10 to 15%	Moderate deficit (8 to 12 weeks maximum)
20 to 30	Significant HPG suppression; low T3; elevated cortisol	Aggressive deficit; risk of RED-S
Below 20	Severe HPG suppression; amenorrhea; near-zero T3	Extreme restriction; immediate intervention required

15.2 Macronutrient Hierarchy for Hormonal Health

The macronutrient hierarchy prioritises: (1) Fat: minimum 0.5 to 0.8 g/kg/day (0.6 g/kg for females) for steroidogenesis. (2) Protein: 1.6 to 2.4 g/kg/day. (3) Carbohydrates: minimum 100 to 150 g/day for most athletes; periodised around training. Very-low-carb diets can elevate cortisol and suppress T3 independent of total calorie intake.

15.3 Key Micronutrients

Table 15.2 Micronutrients for Hormonal Health

MICRONUTRIENT	ROLE	FOOD SOURCES	SUPPLEMENTAL DOSE
Zinc	Steroidogenesis cofactor; 5a-reductase; aromatase inhibition	Oysters, beef, pumpkin seeds	15 to 30 mg/day

Magnesium	300+ enzymes; steroidogenesis; LH sensitivity; sleep	Pumpkin seeds, spinach, dark chocolate	200 to 400 mg/day
Vitamin D	VDR in Leydig cells; upregulates steroidogenic enzymes	Fatty fish, egg yolk	1,000 to 3,000 IU/day
Boron	Reduces SHBG, increases free T	Avocado, raisins, almonds	3 to 6 mg/day
Iodine	Required for T3/T4 synthesis	Seaweed, iodised salt, fish	150 mcg/day
Selenium	Deiodinase cofactor (T4 to T3)	Brazil nuts, tuna, eggs	55 to 200 mcg/day

15.4 Meal Timing

Frequent feeding (six or more meals per day) suppresses lipolysis by keeping insulin elevated. Time-restricted feeding extends the low-insulin window but must not compromise total energy intake or training performance. Pre-sleep casein (30 to 50 g) supports overnight MPS. Post-training nutrition within 2 to 3 hours shifts the body from a catabolic to an anabolic state.

CHAPTER SUMMARY

Energy availability is the primary nutritional variable. Below 30 kcal/kg FFM/day, the HPG and HPT axes are progressively suppressed. Hierarchy: EA above fat minimum above protein above carbohydrate periodisation above micronutrients above meal timing.

◈ MAKE THIS WORK FOR YOU

Energy availability (EA) — calories left for bodily function after training — is the master nutritional variable for hormones. **Below ~30 kcal per kg of fat-free mass per day, the reproductive and thyroid axes progressively shut down** — in men and women, athlete and amateur. Almost every "mystery" hormonal problem in the gym is energy availability first.

Quick EA check (no maths degree required): 30 kcal/kg FFM/day ≈ eating at roughly maintenance with a small-to-moderate deficit at most, while training. Crash diets (very low calories + training) push EA well below the threshold — that is why aggressive cuts kill libido, cycles, and thyroid output (Chapter 5). The practical rule: deficits of 300–500 kcal are hormonal-safe for most people; deficits of 800–1000+ kcal with training are not — regardless of how much protein you eat.

If you are dieting: The hormonal safeguards: (1) Keep the deficit modest (300–500 kcal). (2) Take 1–2 week diet breaks at maintenance every 8–12 weeks (restores T3, T, and cycle). (3) Keep fat ≥ 0.6–0.9 g/kg/day (very-low-fat diets suppress T directly — the "fat minimum" in the hierarchy). (4) Keep protein at 1.8–2.2 g/kg. (5) If you are female and your cycle becomes irregular or stops, the deficit is too aggressive — period. Add calories back; the fat loss can resume later from a healthier state.

If you are a serious athlete: Your training burns the energy that would otherwise fuel the axes — this is exactly why LEA and RED-S are endemic in endurance and aesthetic sports (Chapter 22). The athlete's rule: periodise energy availability — eat at maintenance/surplus around hard blocks, allow the deficit only in defined cut phases, and monitor the subjective markers of Chapter 12 (libido, cycle, sleep, mood) as your EA alarm system. Hierarchy reminder: energy availability > fat minimum > protein > carb periodisation > micronutrients > meal timing. Fix in that order, always.

16. Stress, HPA Axis Dysregulation and Allostatic Load

16.1 Defining Allostatic Load

Allostatic load is the cumulative physiological cost of repeated or chronic stress exposures. It is quantified through cortisol (elevated or flattened rhythm), DHEA-S (low), inflammatory markers (IL-6, CRP), blood pressure, and metabolic markers (HbA1c, waist-to-hip ratio). High allostatic load predicts poor training recovery and impaired body composition response.

16.2 The Four Allostatic Load Patterns

Table 16.1 Allostatic Load Patterns

TYPE	PATTERN	EXAMPLE	COACHING IMPLICATION
Repeated hits	Multiple stressors without recovery	Back-to-back hard sessions	Plan proactive deloads
Lack of adaptation	Failure to habituate to repeated stressor	Persistent cortisol to same stimulus	Reduce variability; improve recovery
Prolonged response	Stress response extends beyond stressor	Cortisol stays elevated after training	Add post-training parasympathetic activation
Inadequate response	Blunted cortisol response	No acute rise to heavy session	Reduce total load; increase EA

16.3 Stress Management Interventions

Evidence-based interventions to manage stress include: (1) Slow-paced breathing at 5 to 6 breaths per minute activates parasympathetic tone within 5 minutes. (2) Morning sunlight (10 to 30 minutes within 30 minutes of waking) entrains the cortisol rhythm. (3) Nature exposure (20+ minutes) reduces cortisol. (4) Social connection buffers the cortisol response. (5) Cognitive diffusion (journaling, mindfulness) reduces psychological stress impact. (6) Environmental modifications reduce baseline sympathetic activation.

CHAPTER SUMMARY

Allostatic load is the cumulative cost of stress. Four patterns describe its manifestation. Screening tools such as the PSS-10 and daily readiness ratings identify accumulating stress. Six interventions reduce allostatic load. The coach must treat stress as a training variable and programme recovery accordingly.

◈ MAKE THIS WORK FOR YOU

Allostatic load is the total stress bill — training, work, family, money, sleep debt, and worry, summed up. When the bill exceeds your capacity, the hormonal system pays: cortisol stays elevated, testosterone and thyroid output drop, and recovery dies. **Stress is a training variable** — it deserves the same respect as volume and intensity, because it uses the same recovery budget.

If you are a high-stress professional who trains: Your training capacity is reduced by your life stress, whether you accept it or not. The practical protocol: (1) Measure it — a daily 1–10 stress score plus the PSS-10 questionnaire (10 questions, free) quarterly gives you a number. (2) Programme around it — hard sessions on low-stress days, lighter/technique sessions on high-stress days (the readiness scoring of recovery book Chapter 6/22 is your decision tool). (3) Manage the load — the six evidence-based load-reducers: sleep extension, morning light, breathing practice (10 min/day), nature walks, reduced alcohol, and deliberate wind-downs. (4) Accept the trade-off: during a genuinely heavy life phase, maintenance training (strength book Chapter 23) is strategic, not weak.

If you are a coach: Your first question for a stalled client is not "what's your programme?" — it is "what else is happening in your life?" A client in a divorce, a job change, or with a sick parent will not respond to programme changes; they need load reduction, sleep support, and realistic expectations. The four allostatic patterns in this chapter (acute spike, chronic elevation, exhaustion, masked) map to different interventions — review them when a client stalls "for no reason".

The key warning: Chronic stress symptoms (fatigue, poor sleep, low libido, weight gain, anxiety) mirror hormonal disease symptoms — but the treatment is load reduction, not hormone replacement. Only when load is managed and symptoms persist 4–8 weeks later do you escalate to labs (Chapter 20). This ordering protects you from the two most common mistakes: medicating a lifestyle problem, or ignoring a real disease.

17. Endocrine Disruptors and Environmental Toxins

17.1 Endocrine-Disrupting Chemicals

EDCs are pervasive in the modern environment: BPA/BPS (plastics), phthalates (fragrances), PFAS (non-stick cookware), parabens (personal care), pesticides (conventional produce), and heavy metals. They interfere with hormone synthesis, receptor binding, and metabolism, creating a low-level chronic hormonal burden that amplifies the effects of poor lifestyle habits.

17.2 Practical EDC Reduction Protocol

The following protocol reduces EDC burden meaningfully without requiring perfection: (1) Water filtration, activated carbon removes most organic EDCs; reverse osmosis removes PFAS. (2) Never heat food in plastic; use glass or stainless steel containers. (3) Choose organic for thin-skinned produce (the Dirty Dozen). (4) Use fragrance-free, paraben-free personal care products. (5) Replace non-stick cookware with cast iron or stainless steel. (6) Use a HEPA filter with activated carbon in the bedroom. (7) Support detoxification pathways with sulphoraphane (broccoli sprouts), cruciferous vegetables (DIM, I3C), adequate fibre, and regular sweating.

CHAPTER SUMMARY

EDCs interfere with hormone function at multiple levels. Complete avoidance is impractical, but meaningful reduction is achievable through water filtration, glass food storage, organic produce, non-toxic personal care and cookware, air filtration, and supporting detoxification pathways.

◈ MAKE THIS WORK FOR YOU

Endocrine-disrupting chemicals (EDCs — BPA/plastics, PFAS, phthalates, pesticides) are real, ubiquitous, and relevant at the margins of hormonal health. The honest framing: **complete avoidance is impossible; meaningful reduction is achievable and worth doing in priority order** — without letting the topic become a fear-driven supplement industry. The "detox" supplement industry is largely marketing; your liver, kidneys, and sweat glands are the detox system, and the best support for them is hydration, vegetables, fibre, and not overloading them.

The reduction checklist (highest value first):

- **Plastics and heat:** Never microwave in plastic (heat leaches BPA/phthalates) — use glass or ceramic. Stop buying water in single-use plastic when tap water is an option; a simple carbon filter (jug or fitted) removes many common contaminants at low cost.
- **Food storage:** Replace plastic food containers and cling film with glass, especially for hot or fatty foods (fat solubilises leached compounds).
- **Cookware:** Non-stick pans over 200–230°C can release PFAS — use stainless steel, cast iron, or ceramic for high-heat cooking; replace scratched non-stick (scratches = more release).
- **Personal care:** "Fragrance" on labels hides phthalates — choose fragrance-free or EWG-verified products for deodorant, shampoo, and laundry where cost allows.
- **Produce:** If budget allows, prioritise organic for the highest-pesticide crops (the EWG Dirty Dozen list is a practical guide) — but always: eating conventional produce beats eating no produce.

If you are trying to conceive: This is where EDC reduction matters most — couple-level exposure reduction (plastics, personal care, pesticides) in the 3–6 months before trying is the evidence-supported window. The same checklist applies, with extra discipline. This is medical-adjacent territory — your fertility specialist is the right person for the details.

If you are otherwise healthy: Do the high-value items (microwave glass, water filter, glass storage) and move on — the marginal benefit of extreme measures (specialty "detox" diets, air-purifier systems everywhere, organic-everything) is small relative to the basics: sleep, energy availability, stress, training. Keep perspective: lifestyle levers (Chapters 13–16) are worth 10x the EDC measures for the average person.

18. Circadian Biology and Hormonal Timing

18.1 The Master Clock

The suprachiasmatic nucleus (SCN) is the master circadian clock, entrained by light via the retinohypothalamic tract. Every cell in the body has its own clock driven by CLOCK/BMAL1 and PER/CRY feedback loops. This system coordinates the timing of hormone release, enzyme activity, and metabolic pathways.

18.2 Circadian Hormonal Timing

Table 18.1 Circadian Timing of Key Hormones

HORMONE	PEAK	TROUGH	COACHING RELEVANCE
Cortisol	6 to 8 AM (CAR)	Midnight	Schedule high-intensity training when cortisol is naturally elevated (AM)
Melatonin	2 to 4 AM	Noon	Blue light after sunset suppresses melatonin and delays sleep onset
Testosterone	6 to 8 AM	8 to 10 PM	25 to 30% higher in AM; labs must be drawn in the morning
GH	11 PM to 3 AM (SWS)	Afternoon	Sleeping before midnight maximises the nocturnal GH pulse
Insulin sensitivity	Morning (highest)	Evening (lowest)	Carbohydrates are better tolerated earlier in the day

18.3 Circadian Optimization Protocol

Circadian Health Protocol

1. **Morning light:** 10 to 30 minutes of outdoor light within 30 minutes of waking.
2. **Consistent sleep timing:** Within 30 to 60 minutes every day, including weekends.
3. **Light after sunset:** Dim lights; use blue-blocking glasses 60 to 90 minutes before bed.
4. **Temperature rhythm:** Cool bedroom (18 to 21 degrees C) at night; bright light and movement in the morning.
5. **Feeding window:** Consistent meal timing; avoid eating 2 to 3 hours before bed.

6. **Training timing:** Morning training reinforces the cortisol rhythm. Consistency matters more than absolute timing.

CHAPTER SUMMARY

The SCN coordinates hormone timing to the 24-hour light-dark cycle. Circadian disruption impairs fat oxidation, glucose tolerance, and appetite regulation. The protocol emphasises morning light exposure, consistent sleep timing, evening light management, and a structured feeding window.

◈ MAKE THIS WORK FOR YOU

Every hormone in this book runs on the same 24-hour schedule — testosterone peaks in the morning, cortisol wakes you, melatonin closes the day, GH pulses at night. **Timing IS the mechanism**: disrupting the rhythm disrupts the hormones. This chapter is the circadian translation of the sleep book's protocols, focused on hormonal output.

The four-tool circadian protocol:

- **Morning light (10–30 minutes, within an hour of waking):** The anchor — it sets the cortisol peak, reinforces the testosterone rhythm, and improves that night's sleep. Highest value when you are stuck inside all day.
- **Consistent sleep timing:** The same wake time 7 days/week is the strongest single anchor (sleep book Chapter 1). Irregularity is circadian jet lag, with the same hormonal costs as the real thing.
- **Evening light management:** Dim lights and no screens 30–60 minutes before bed protect melatonin — the hormone that gates everything else (sleep book Chapter 10).
- **Structured feeding window:** Eating within a consistent daytime window (breakfast to early dinner), with the overnight fast preserved, supports insulin sensitivity (Chapter 6) and overnight GH pulsatility. No midnight snacking — it is the cheapest hormonal timing fix there is.

If you are a shift worker: Your hormones run on a foreign clock — the protocol is the same tools, shifted (sleep book Chapter 21): anchor to your shift, bright light during your work "day", strict dark + wind-down after, meals relative to your shift, and planned recovery sleeps between rotations. Expect some hormonal cost (it is documented); the protocol limits it. If symptoms (sleep, mood, weight) persist despite running it well for months, have the schedule conversation with your employer and the health conversation with your doctor — this is one of the few cases where the job itself can be the health problem.

If you are a night owl in a day world: Morning light is your best tool for pulling the rhythm as early as your biology allows (sleep book Chapter 15); failing that, protect consistency — a stable late rhythm beats a swinging one. And never "compensate" with late caffeine (it delays the melatonin rhythm further — sleep book Chapter 17).

19. Supplements for Hormonal Optimization

19.1 Tier 1: High-Confidence Supplements

Table 19.1 High-Confidence Hormonal Support Supplements

SUPPLEMENT	MECHANISM	DOSE	EVIDENCE LEVEL
Vitamin D3	VDR activation in steroidogenic tissues	1,000 to 3,000 IU/day	Strong (RCTs show T rise in deficient)
Zinc	Steroidogenesis cofactor; aromatase inhibition	15 to 30 mg/day	Strong (deficiency correction raises T)
Magnesium glycinate	Steroidogenesis; sleep quality; LH receptor sensitivity	200 to 400 mg before bed	Moderate to Strong
Creatine monohydrate	Improves training output, supporting hormonal adaptation	3 to 5 g/day	Strong (indirect hormonal benefit)
Omega-3 (EPA/DHA)	Reduces inflammation; supports LH receptor sensitivity	1.5 to 4 g/day	Moderate

19.2 Tier 2: Conditionally Effective

Table 19.2 Conditionally Effective Supplements

SUPPLEMENT	DOSE	BEST FOR	EVIDENCE LEVEL
Ashwagandha	300 to 600 mg/day	High stress; elevated cortisol	Moderate (cortisol reduction in stressed)
D-Aspartic Acid	2,000 to 3,000 mg/day (12 to 28 days only)	Short-term LH pulse augmentation	Moderate (effect diminishes after 2 to 4 weeks)
Boron	3 to 6 mg/day	Elevated SHBG; low free T	Moderate (SHBG reduction)

19.3 Supplements Requiring Medical Supervision

SARMs, prohormones, aromatase inhibitors, SERMs (clomiphene, tamoxifen), and high-dose DHEA (above 50 mg/day) should never be recommended or used without medical oversight.

Important Caution

Supplements are adjuncts, not substitutes for lifestyle optimisation. No supplement can overcome the hormonal impact of less than six hours of sleep, prolonged caloric deficit, unmanaged chronic stress, or visceral adiposity. The correct hierarchy is: lifestyle first, micronutrient correction second, evidence-based supplementation third.

CHAPTER SUMMARY

Vitamin D, zinc, and magnesium are high-confidence supplements for hormonal support. Ashwagandha is conditionally effective for stress-related suppression. Boron may reduce SHBG. SARMs and prohormones require medical supervision. Supplements are adjuncts, not alternatives, to lifestyle optimisation.

◈ MAKE THIS WORK FOR YOU

The honest supplement framework for hormones: three high-confidence basics, a couple of conditional options, and a long list of marketing. **Supplements correct deficiencies and support lifestyle; they do not replace it.** The hierarchy always stands: sleep → energy availability → stress → training → supplements → labs.

The high-confidence trio (worth doing for most people):

- **Vitamin D (test first — 1,000–4,000 IU/day as needed):** Deficiency is extremely common and directly linked to testosterone levels, mood, and bone health. This is the supplement with the most robust hormonal evidence — and it only works if you are deficient (hence: test).
- **Zinc (15–30 mg/day):** Deficiency (common in heavy sweaters and vegetarians) suppresses T; repletion restores it. Zinc only helps if deficient — and note zinc competes with copper, so high-dose long-term use needs balance.
- **Magnesium (200–400 mg/day, evening):** Supports sleep, stress response, and recovery; deficiency is common. Glycinate form for sleep benefits.

Conditional options: Ashwagandha (300–600 mg/day of standardised extract) has the strongest RCT evidence of any herbal for reducing stress and cortisol — effective for stress-elevated cortisol, useless for normal levels; N-of-1 test it for 8–12 weeks (it is a supplement experiment, not a promise). Boron (3–6 mg/day) has small studies showing SHBG reduction and free-T increase — modest, individual, and safe at that dose; worth a trial if free T is low with normal total T (Chapter 8). Fish oil supports overall inflammatory and hormonal health as part of the foundation.

Avoid these: "Natural testosterone boosters" (every formula on the market — the ingredients with any evidence are the ones above, at doses too small to matter); SARMs and prohormones (medical-supervision-only compounds with real side effects — buying them from supplement sites is how people damage their HPG axis, Chapter 2); high-dose anything without a deficiency; and adaptogen blends with 12 ingredients (uninterpretable, unproven).

The one-sentence rule: Test for deficiencies, fix the lifestyle, and use supplements to fill measured gaps — not to purchase hope. If you have been supplementing for 3 months with no measurable change in symptoms or labs, stop and redirect the money to sleep or food.

Coaching Application

Domain 3 Case Study: The Insomniac Executive

Scenario: A 42-year-old male executive, 84 kg, 14% body fat, has seen progress stall over 6 weeks. Sleep has deteriorated to 5 to 5.5 hours/night, morning erections and libido have declined. He is eating at maintenance (2,600 kcal/day), training 5x/week, under high work stress. Labs: total T 380 ng/dL (down from 520 ng/dL six months ago), SHBG 52 nmol/L, free T 48 pg/mL, cortisol (evening) 18 mcg/dL.

Assessment: Sleep deprivation is the primary driver. The decline from 520 to 380 ng/dL with unchanged nutrition and training points to a lifestyle variable, and that variable is sleep. Elevated evening cortisol confirms HPA disruption.

Intervention: (1) Sleep protocol targeting 7.5+ hours per night. (2) Reduce training to 4x/week at RPE 7. (3) Magnesium glycinate 400 mg plus ashwagandha 600 mg. (4) Caffeine cutoff at 12 PM. (5) Reassess sleep quality in 2 weeks; repeat labs in 8 weeks.

Key Takeaway: When a previously progressing client plateaus with concurrent sleep disruption, the cause is almost always a lifestyle variable and sleep is the highest-leverage intervention. Training volume must be reduced during high-stress periods, not increased.

DOMAIN 4

Applied Protocols and Clinical Integration: lab interpretation, sex-specific protocols, contraception, age-related decline, and complete coaching assessment

20. Hormonal Assessment and Lab Interpretation

20.1 When to Order Labs

Not every coaching client requires blood work, but clear indications exist: (1) unexplained plateau in body composition progress lasting 8+ weeks despite adherence, (2) symptoms of hormonal dysfunction (low libido, chronic fatigue, menstrual irregularity, poor recovery), (3) age over 40 for baseline screening, (4) history of endocrine conditions (thyroid disorders, PCOS, diabetes), (5) prolonged caloric restriction beyond 12 weeks, and (6) athletes

Free testosterone (or calculated)	Bioavailable androgen fraction	More clinically relevant than total; accounts for SHBG variation
SHBG	Binding protein status	Determines free fraction; modulated by insulin, E2, T3
LH and FSH	Pituitary gonadotropin function	Differentiates primary vs. secondary hypogonadism
Prolactin	Dopaminergic tone	Elevated prolactin suppresses HPG axis
Cortisol (AM)	HPA axis function	Elevated or flattened rhythm indicates allostatic load
TSH, free T4, free T3	Thyroid axis	Identifies subclinical, primary, or low T3 syndrome
Reverse T3	Peripheral thyroid conversion	Elevated rT3 with low T3 confirms low T3 syndrome
Estradiol (E2)	Aromatase activity	High E2 in males indicates excess aromatisation
IGF-1	GH axis function	Marker of anabolic signalling capacity
Ferritin	Iron stores	Iron deficiency impairs thyroid function and energy metabolism
Vitamin D	Steroidogenic cofactor	Deficiency is common and directly impacts hormone synthesis

20.3 Lab Interpretation Framework

Lab interpretation follows a systematic process. Start with the thyroid axis (TSH, T4, T3, rT3) to determine whether metabolic rate is normal, suppressed, or elevated. Then assess the HPG axis: total T, free T, LH, FSH, prolactin, and E2. The pattern of LH/FSH relative to total T tells you whether the dysfunction is central (pituitary/hypothalamus) or peripheral (gonads). Next, examine cortisol rhythm and the C:T ratio. Finally, evaluate the metabolic context: energy availability, sleep, stress, body composition, and training load. Labs without context are misleading, a low testosterone in the setting of a 16-week deficit is expected physiology, not pathology.

> **Clinical Referral Thresholds**

> The following findings warrant referral to a physician: total testosterone consistently below 250 ng/dL with symptoms, TSH above 10 mIU/L, prolactin above 50 ng/mL, haemoglobin A1c above 6.5%, or any flagrantly abnormal value outside the reference range with corresponding symptoms. The coaches scope is lifestyle optimisation, not medical treatment.

20.4 The Optimal vs. Clinical Distinction

Clinical reference ranges are population-based and include individuals with varying degrees of dysfunction. The optimal range for performance and well-being is narrower. A total testosterone of 350 ng/dL is within the clinical reference range but is suboptimal for muscle gain, libido, energy, and recovery in most men. The coach works in the gap between not clinically deficient and optimal. This gap is where lifestyle interventions have the greatest impact.

CHAPTER SUMMARY

Hormonal assessment should be triggered by clear clinical flags: plateaus, symptoms, age, history, and prolonged restriction. The minimal panel covers the HPG, HPA, and HPT axes plus key modulators (SHBG, prolactin, ferritin, vitamin D). Lab interpretation must be contextualised with lifestyle variables. The coach operates in the gap between clinical deficiency and optimal function.

◈ MAKE THIS WORK FOR YOU

Testing is only useful when you know **when to test, what to test, and how to interpret** — this chapter is that manual. The three mistakes people make: testing without clinical flags (waste of money), testing the wrong panel (total T only — Chapter 8), and interpreting numbers without lifestyle context (Chapter 16's "medicating a lifestyle problem").

When to test (the five triggers): (1) Plateaus that resist programme changes for 8+ weeks across all lifts. (2) Symptoms — libido, energy, mood, cycle changes lasting 4+ weeks with clean lifestyle. (3) Age — routine screening from ~45–50 for men (andrologists recommend conversation-based screening; the test is your data). (4) History — known thyroid, diabetes, or pituitary issues in the family. (5) Prolonged restriction — after 6+ months of aggressive dieting, a check is reasonable. If none of these apply, the test can wait.

The minimal panel (what to actually order): HPG: total T, free T (calculated or direct), SHBG, LH, FSH, estradiol (sensitive assay for men), prolactin. HPA: morning cortisol (8–9am), sometimes DHEA-S. HPT: TSH, free T4, free T3. Modulators: ferritin, vitamin D, fasting glucose/HbA1c. Standardise: morning, fasted, rested (no training 24–48h before), same lab — so results are comparable over time.

How to interpret (the coach's frame): Labs have two layers: clinical deficiency (disease — your clinician's territory) and optimal function (the gap where coaching lives). Most athletes will be "in range" but below their personal optimum — that is the coaching gap, and the intervention is the lifestyle levers of Chapters 13–19, not medication. Compare against *your* history, not just the reference range (a drop from 700 to 450 with symptoms matters, even if 450 is "normal"). And always read labs with the lifestyle story: same numbers with 5h sleep versus 8h sleep are different problems.

21. Male Hormonal Optimization Protocol

21.1 The Four-Phase Optimization Model

Male hormonal optimization follows a four-phase hierarchical model: Phase 1 addresses lifestyle fundamentals, Phase 2 targets micronutrient optimisation, Phase 3 applies targeted supplementation, and Phase 4 involves clinical referral for persistent dysfunction. Each phase builds on the previous one, and moving to the next phase requires documented failure of the prior phase after adequate trial duration.

21.2 Phase 1: Lifestyle Foundation (Weeks 1 to 12)

Table 21.1 Phase 1 Lifestyle Intervention Targets

DOMAIN	TARGET	MONITORING
Energy availability	Above 30 kcal/kg FFM/day; no deficit beyond 12 weeks without break	Weekly weight trend; dietary log review
Dietary fat	0.6 to 1.0 g/kg/day; at least 25% of total calories	Fat intake tracking; fatty acid profile
Sleep	7 to 9 hours; consistent bedtime within 30 min	Sleep diary or wearable data
Body fat	Reduce if above 20%; optimise visceral adiposity	Waist circumference; DEXA trend
Training volume	10 to 20 sets/muscle/week; 60 to 75 min sessions	RPE tracking; performance trend
Stress management	Daily downregulation practice; PSS-10 below 15	Subjective stress rating; HRV trend

21.3 Phase 2: Micronutrient Correction (Weeks 1 to 12, concurrent)

Correct deficiencies through diet and targeted supplementation: (1) Vitamin D, target serum 50 to 80 ng/mL through supplementation (1,000 to 3,000 IU/day or as guided by labs). (2) Zinc, ensure sufficiency through diet (oysters, beef, pumpkin seeds) or 15 to 30 mg/day supplementation. (3) Magnesium, 200 to 400 mg magnesium glycinate before bed supports both steroidogenesis and sleep. (4) Assess iron status via ferritin; ferritin below 30 ng/mL impairs thyroid function and energy metabolism in at-risk populations.

21.4 Phase 3: Targeted Supplementation (Weeks 8 to 12, conditional)

If free testosterone remains below 80 pg/mL (male) after 8 weeks of Phase 1 and 2 adherence, consider: (1) Boron 3 to 6 mg/day to reduce SHBG and increase free T. (2) Ashwagandha 600 mg/day if cortisol is elevated or perceived stress is high. (3) D-Aspartic Acid 2,000 to 3,000 mg/day for 12 to 28 days only (tachyphylaxis develops rapidly). These supplements work best when the lifestyle foundation is solid; they cannot compensate for poor sleep or chronic energy deficit.

> **Important Note**
>
> No supplement or lifestyle intervention will restore testosterone to youthful levels in the presence of primary hypogonadism or significant testicular pathology. The optimization protocol applies to functional (secondary) hypogonadism only. If total testosterone remains below 250 ng/dL despite 12 weeks of optimal lifestyle implementation, medical referral for TRT consideration is appropriate.

21.5 Phase 4: Clinical Referral

If total testosterone is below 250 ng/dL with symptoms, or if free testosterone remains below 65 pg/mL despite 12 weeks of comprehensive lifestyle and supplementation intervention, refer the client to a physician specialising in mens health. The coach continues to support lifestyle optimisation alongside any medical intervention. TRT is a medical decision with significant implications for fertility, cardiovascular health, and haematocrit; it requires ongoing medical supervision.

CHAPTER SUMMARY

The male hormonal optimization protocol follows four phases: lifestyle foundation, micronutrient correction, targeted supplementation, and clinical referral. Each phase builds on the prior one. Most functional hypogonadism responds to Phase 1 and 2 interventions alone. The protocol emphasises energy availability, sleep, dietary fat, body composition, and stress management as the primary levers.

◈ MAKE THIS WORK FOR YOU

This is the practical protocol: four phases, each building on the last, each taking 8–12 weeks. The key insight: **most men with "low T feelings" resolve in Phases 1–2 — before any supplement or prescription**. Work the phases in order; skipping ahead is how people waste money on supplements or jump to TRT prematurely.

Phase 1 — Lifestyle foundation (weeks 1–12): Sleep 7–9 hours, consistent timing (Chapter 13 — the single biggest lever). Energy availability at maintenance or a modest deficit (no crash diets — Chapter 15). Dietary fat ≥ 0.6–0.9 g/kg/day. Body fat: work toward a healthy waist (< 94 cm) — the aromatase loop reverses with weight loss (Chapter 9). Stress managed (Chapter 16), alcohol minimal. Training: 3–5 sessions/week with heavy compounds, deloads honoured (Chapter 14). Track: libido, energy, sleep, strength.

Phase 2 — Micronutrient correction (weeks 4–12, can overlap): Test vitamin D, ferritin, zinc status; correct measured deficiencies (Chapter 19's trio: vitamin D, zinc, magnesium). Test first — supplements only fill measured gaps.

Phase 3 — Targeted supplementation (weeks 12–24, only if needed): Only if Phases 1–2 have not resolved symptoms: ashwagandha for stress-pattern (high cortisol symptoms), boron if free T is low with high SHBG (Chapter 8). N-of-1 test each for 8–12 weeks with measurable outcomes; discard what does not work.

Phase 4 — Clinical referral (when warranted): If after 6 months of clean Phases 1–3 you still have symptoms *and* low morning total/free T with abnormal LH/FSH or prolactin (Chapter 10), the referral is the correct, efficient move — TRT or treatable causes (prolactin, thyroid, etc.) are the physician's domain. Track record for the doctor: your lifestyle log, your panel history, your N-of-1 results. A well-documented referral gets better care than a vague one.

22. Female Hormonal Optimization Protocol

22.1 The Female-Endocrine Distinction

Female hormonal optimization differs from male in several critical respects. Energy availability thresholds are higher. The menstrual cycle creates fluctuating hormonal backgrounds that require a more nuanced approach to programming and nutrition. Estrogen is the primary anabolic sex hormone, and its cyclic variation is normal and adaptive, not something to be suppressed or flattened. The goal is to support healthy cycle function, not to create a constant hormonal state.

22.2 Phase 1: Cycle Health Foundation

Table 22.1 Female Hormonal Optimization: Lifestyle Targets

DOMAIN	TARGET	SPECIAL CONSIDERATION
Energy availability	Above 35 to 40 kcal/kg FFM/day	Higher threshold than males; preserve menstrual function
Dietary fat	0.7 to 1.0 g/kg/day	Fat is essential for E2 and progesterone synthesis
Carbohydrates	150 to 250 g/day minimum	Low-carb diets suppress T3 and elevate cortisol in women
Sleep	7 to 9 hours; consistent timing	Sleep disruption affects cycle regularity
Training	Periodised; reduce volume in luteal phase	Luteal phase: RPE 6 to 8; follicular phase: RPE 7 to 9
Stress management	Daily practice; HRV monitoring	Females show greater cortisol reactivity to psychological stress

22.3 Cycle-Phase Programming

Training and nutrition should be adjusted across the cycle without rigidly over-prescribing. During the follicular phase (days 1 to 13), estrogen rises, insulin sensitivity is higher, and the anabolic environment is favourable. This is the window for higher-intensity training, progressive overload, and higher carbohydrate intake. During the luteal phase (days 15 to 28), progesterone rises, catabolic signalling increases, and exercise tolerance decreases. Training volume and intensity should be reduced, protein intake emphasised, and carbohydrate intake maintained or slightly increased to support serotonin synthesis and mood regulation.

22.4 RED-S Prevention and Management

Relative Energy Deficiency in Sport (RED-S) is underdiagnosed in female athletes and active women. Key indicators: menstrual irregularity (cycles shorter than 21 days or longer than 35 days), history of stress fractures, low energy availability, disordered eating patterns, and recurrent illness or injury. Management requires increasing energy availability to restore menstrual function, which may require a sustained period at maintenance or surplus calories. Bone mineral density lost during prolonged RED-S is partially but not fully recoverable, making early intervention critical.

CHAPTER SUMMARY

Female hormonal optimization requires higher energy availability, adequate fat and carbohydrate intake, cycle-phase programming, and vigilant RED-S prevention. The goal is to support healthy menstrual function as the marker of a balanced hormonal milieu. Programming should acknowledge cycle phase without rigid over-prescription. Early identification of RED-S is critical for preserving bone health.

◈ MAKE THIS WORK FOR YOU

The female protocol's goal is simple and specific: **a healthy menstrual cycle is the report card** — it only functions when energy, sleep, and stress are adequate. If the cycle is regular, the hormonal environment is largely working. If it is not, that is the first thing to fix, and it overrides every training and nutrition goal.

The core protocol: (1) Energy availability at or above ~40 kcal/kg FFM/day — female athletes need higher EA than males; deficits that men tolerate silently suppress the female cycle (Chapter 15 — the threshold is higher for women). (2) Adequate fat (≥ 0.9–1.0 g/kg) and carbohydrates around training (the female HPT and HPG axes are carb-sensitive). (3) Sleep 7–9 hours (Chapter 13 — women lose more sleep to stress than men on average; protect it). (4) Cycle-aware programming, not rigid: track 2–3 cycles, adjust only if you observe real patterns (Chapter 7 — heavier work in the follicular phase, gentler expectations in the late luteal). (5) Iron status tested annually — ferritin ≥ 30 ng/mL is the athletic target; deficiency is common, treatable, and quietly destroys performance.

RED-S awareness (the critical flag): Relative Energy Deficiency in Sport — a lost/irregular cycle + low energy + heavy training = RED-S until proven otherwise. It is not "a training break" — it is a bone-health emergency risk (bone density is lost and not fully recoverable). The response: raise energy availability immediately, reduce training load, involve a clinician. There is no "train through it" version of RED-S. Chapter 12's subjective markers (cycle, libido, sleep, mood) are your early-warning system — use them.

If you are perimenopausal/menopausal: The protocol shifts (Chapter 11): cycle-based programming ends, resistance training becomes the priority, protein rises, and the sleep toolkit (night sweats — sleep book Chapters 9 and 25) is essential. HRT conversations with a menopause-informed clinician are strongly worth having — optimised hormones at this stage protect bone, muscle, and cognition for decades.

23. Hormonal Contraception and Athlete Management

23.1 Types of Hormonal Contraception

Table 23.1 Hormonal Contraception Types and Endocrine Effects

TYPE	HORMONES	EFFECT ON ENDOGENOUS CYCLE	COACHING IMPLICATIONS
Combined oral contraceptive	Ethinyl estradiol + progestin	Suppresses HPG axis; no ovulatory cycle	No cycle phase variation; constant hormonal state
Progestin-only (POP/minipill)	Progestin only	Variable HPG suppression	More individual variation; unpredictable response
IUD (hormonal)	Levonorgestrel	Local uterine effect; ovulation often preserved	Minimal systemic impact; cycle may continue
Implant (Nexplanon)	Etonogestrel	Suppresses ovulation in most users	Constant progestin exposure; androgenic effects vary
Injection (Depo-Provera)	Medroxyprogesterone acetate	Suppresses HPG axis; prolonged effect	Associated with bone density loss; long washout period

23.2 How Contraception Alters the Coaching Landscape

Hormonal contraception eliminates the natural menstrual cycle, replacing cyclic hormonal variation with a constant (or stepped) exogenous hormone profile. Cycle-phase programming becomes irrelevant because there is no endogenous cycle to track. COCs reduce free testosterone by increasing SHBG (ethinyl estradiol is a potent SHBG upregulator), which may blunt the anabolic response to training in some women. Progestin-only methods vary widely in their androgenic profile and metabolic effects.

23.3 Contraception, Body Composition, and Performance

Research on contraception and body composition outcomes in athletes is limited but points to several considerations: COC users may experience reduced MPS compared to naturally cycling women, potentially due to lower free testosterone and IGF-1. Depot medroxyprogesterone acetate is associated with weight gain and reduced bone mineral density with prolonged use. Androgenic progestins (levonorgestrel, norethisterone) may have neutral or slightly positive effects on lean mass compared to anti-androgenic progestins (drospirenone).

23.4 Practical Coaching Recommendations

For clients on hormonal contraception: (1) Track symptoms, not cycle phases, the rhythm method for programming does not apply. (2) Monitor libido, recovery, and mood as indicators of tolerance to the specific contraceptive type. (3) If a client experiences negative side effects, discuss the possibility of switching to a different formulation or a non-hormonal method with their physician. (4) For athletes relying on natural cycle tracking for performance, non-hormonal or copper IUD options preserve the endogenous cycle.

CHAPTER SUMMARY

Hormonal contraception replaces the natural menstrual cycle with an exogenous hormonal profile. COCs reduce free testosterone through SHBG upregulation, potentially blunting the anabolic response. Cycle-phase programming is not applicable for COC users. The coach should understand the specific contraceptive and help the client make informed choices.

◈ MAKE THIS WORK FOR YOU

Hormonal contraception matters for training because it **replaces the hormonal cycle with a different profile** — and different methods do different things. The coaching rule: know the method, adjust expectations, and never let "the pill" become a lazy explanation for a client's symptoms.

If you use combined oral contraceptives (COC — the pill): Key facts: (1) Your cycle is artificial — cycle-phase programming (Chapter 7) does not apply; your hormones are flat and consistent, which usually means consistent training performance. (2) COCs raise SHBG 2–3x, binding free testosterone (Chapter 8) — this can slightly reduce the anabolic environment and, for some women, libido. (3) Practical implications: programme consistently without cycle adjustments; if libido or training response is flat, this is a legitimate medical conversation (a different formulation or method may change things) — but not a fitness failure.

If you use hormonal IUDs or progestin-only methods: Local or progestin-only effects — usually minimal impact on training; most women report consistent performance. Treat it as neutral and programme normally.

If you are choosing or switching methods as an athlete: An informed choice conversation includes: performance consistency (COCs flatten the cycle — good for some), bone health (long-term COC use has mixed bone data; non-hormonal and progestin-only options differ), and the specific sport context. This is a doctor conversation — your job is to bring the training context. And note: if you stop the pill, expect a transition period (weeks to months) while your natural cycle returns — performance may fluctuate during it; do not panic or change everything at once.

If you are a coach: Ask about contraceptive method only as health context (with appropriate boundaries) — it legitimately changes cycle-based programming and interpretation of symptoms (Chapter 7's "track your pattern" only applies to natural cycles). Never advise starting, stopping, or switching contraception — that is medical territory. Your value is helping the client interpret what their body does and connecting symptoms to the right professional conversation.

24. Age-Related Hormonal Decline and Interventions

24.1 The Male Aging Endocrine Profile

Testosterone declines at approximately 1 to 2% per year after age 30. This decline is gradual and highly variable between individuals. SHBG rises with age, further reducing free testosterone. GH and IGF-1 decline progressively (the somatopause). Cortisol tends to increase or become dysregulated. Melatonin production declines, contributing to age-related sleep deterioration. These changes collectively reduce the anabolic ceiling, increase catabolic tone, and impair recovery capacity. However, lifestyle factors, particularly body fat, physical activity, and sleep, moderate the rate of decline significantly.

24.2 The Female Aging Endocrine Profile

Table 24.1 Female Hormonal Transition Stages

STAGE	AGE RANGE	KEY HORMONAL CHANGES	BODY COMPOSITION IMPACT
Late reproductive	35 to 40	Subtle E2 decline; rising FSH; shorter cycles	Minimal; maintain resistance training foundation
Early perimenopause	40 to 45	Variable E2; erratic cycles; P4 decline first	Mild increase in visceral fat tendency
Late perimenopause	45 to 50	Significant E2 fluctuations; anovulatory cycles	Accelerated muscle loss risk; sleep disruption
Menopause	Approximately 51	E2 at postmenopausal levels; FSH elevated	Rapid visceral fat gain; accelerated bone loss
Post-menopause	51+	Stable low E2; adrenal androgens dominant	Reduced anabolic ceiling; higher protein requirement

24.3 Lifestyle Interventions for Age-Related Decline

Evidence-based interventions for age-related hormonal decline: (1) Resistance training is the most potent intervention for maintaining muscle mass, bone density, and metabolic health in older adults. It improves androgen receptor sensitivity, GH pulsatility, and insulin sensitivity. (2) Protein intake must increase to 1.8 to 2.4 g/kg/day to overcome anabolic resistance. (3) Vitamin D and calcium become increasingly important for bone health. (4) Sleep quality interventions (bright light therapy, sleep extension, circadian entrainment) can partially reverse age-related hormonal deterioration. (5) Body fat management to maintain visceral adiposity in a healthy range reduces aromatase burden and metabolic inflammation.

24.4 The Role of HRT and TRT

Hormone replacement therapy (HRT for women, TRT for men) is a medical intervention with established benefits and risks. The coaches role is not to recommend or prescribe these therapies but to understand them well enough to work alongside medical providers. Clients on TRT or HRT require adjusted coaching strategies because their hormonal milieu differs substantially from the natural profile. For clients considering these therapies, the coach can help them prepare a list of informed questions for their physician.

CHAPTER SUMMARY

Age-related hormonal decline affects both sexes through reduced anabolic hormones, increased catabolic tone, and impaired recovery. The rate of decline is moderated by lifestyle factors. Resistance training, increased protein, vitamin D, sleep optimisation, and body fat management are the primary interventions. The coaches role is to understand HRT/TRT enough to collaborate with medical providers.

◈ MAKE THIS WORK FOR YOU

Hormones decline with age — but the honest science is encouraging: **much of the decline is lifestyle-caused, and most of it is modifiable**. Men's testosterone drops ~1% per year after 30 (some of that is the fat gain and sleep deterioration that accompany aging — fixable); women face the menopause transition (a real hormonal cliff, with treatments that work). The message for both: aging hormones are a training variable, not a verdict.

If you are a man 40+: The protocol: resistance training 2–4x/week with heavy compounds (the strongest lifestyle stimulator of the anabolic environment at any age — strength book Chapter 21 for the over-40 specifics), protein 1.6–2.2 g/kg (anabolic resistance is real — older muscles need more per meal), vitamin D and zinc status checked, sleep prioritised (sleep quality is one of the biggest age-related decline drivers), and body fat managed (the aromatase loop of Chapter 9 accelerates with age). Expect: you can maintain or even raise your levels with consistent lifestyle — and the reference range itself is not your target; your symptoms and trends are (Chapter 20).

If you are a woman in the menopause transition: The hormonal change is real and life-impacting (sleep, body composition, bone, mood, libido — Chapter 11 has the full picture). The treatments (HRT, and for some, testosterone therapy) are safe and effective for many women and dramatically underused. If symptoms are affecting your training or life, have the conversation with a menopause-informed clinician — this is one of the highest-value medical conversations available to women in their 40s–50s. Alongside: resistance training (non-negotiable — bone and muscle), protein, and the sleep toolkit.

If you are on HRT/TRT: You are a medical patient with an athletic component: monitor with your clinician (regular blood work, symptom review), keep training and nutrition standards high (the therapy supports the environment; the training builds the results), and be honest in logs so your coach and doctor can coordinate. The coach's role is collaboration with the medical provider, not prescription or management — and knowing the difference is part of professional practice (Chapter 25's scope section).

25. Complete Hormonal Health Assessment and Coaching Integration

25.1 The Integrated Assessment Framework

The complete hormonal health assessment integrates four layers: subjective markers (libido, sleep quality, recovery, cycle regularity, mood, energy), anthropometric markers (body fat distribution, waist circumference, muscle mass trends), performance markers (strength progression, recovery rate, HRV trends), and laboratory markers (the minimal panel from Chapter 20). No single layer is sufficient; the integrated picture across all four layers determines the coaching strategy.

Complete Hormonal Health Assessment Protocol

1. **Initial screening:** Administer the hormonal health screening checklist (Chapter 12). Score and identify flags.
2. **Subjective baseline:** Record libido, sleep quality (PSQI or similar), recovery readiness, and stress level (PSS-10).
3. **Anthropometric assessment:** Measure waist circumference, body fat percentage (DEXA preferred, calipers acceptable), and muscle circumference.
4. **Performance audit:** Review training logs for strength progression, recovery between sessions, and HRV trend if available.
5. **Lab review (if available):** Interpret labs using the four-axis framework (thyroid, HPG, HPA, metabolic context).
6. **Lifestyle audit:** Assess energy availability, sleep, stress, training load, nutrition quality, and environmental EDC exposure.
7. **Synthesise:** Identify the primary axis or axes involved. Determine whether the dysfunction is functional (lifestyle-driven) or pathological (requires referral).
8. **Intervention:** Assign the appropriate protocol from Chapters 21 or 22. Schedule reassessment in 8 to 12 weeks.

25.2 The Four-Axis Diagnostic Framework

Table 25.1 The Four-Axis Diagnostic Framework

AXIS	PRIMARY HORMONES	COMMON DYSFUNCTION PATTERN	PRIMARY COACHING LEVERS

HPT (Thyroid)	TSH, T4, T3, rT3	Low T3 syndrome; subclinical hypothyroidism	EA, reverse diet, diet break, carb adequacy
HPG (Reproductive)	Total T, free T, LH, FSH, E2, SHBG	Secondary hypogonadism; low free T; high E2:T	EA, fat intake, sleep, body fat reduction
HPA (Stress)	Cortisol (AM/PM), DHEA-S	Elevated cortisol; flattened rhythm; HPA hypoactivity	Sleep, stress management, training volume, carb timing
Metabolic	IGF-1, insulin, glucose, HbA1c	Insulin resistance; low IGF-1	Training, carb periodisation, fibre, body fat reduction

25.3 The Coaching Integration Matrix

The coaching integration matrix maps each hormonal axis to the corresponding coaching variables. A dysfunction in the HPG axis directs the coach to examine energy availability, dietary fat intake, sleep quality, and body composition. A dysfunction in the HPA axis directs attention to stress management, training volume, and recovery protocols. When multiple axes are involved, the most common presentation, the coach addresses the most upstream axis first: typically, the HPA axis, because elevated cortisol suppresses both the HPG and HPT axes.

Key Point: When multiple hormonal axes are dysregulated, address the HPA axis first. Stress management, sleep, and allostatic load reduction restore the foundation upon which the HPG and HPT axes can recover.

25.4 Putting It All Together: The Coaching Cycle

The hormonal coaching cycle operates on an 8 to 12 week assessment loop. At each reassessment point, the coach reviews subjective markers, objective data, and labs (every second or third cycle). The goal is not to achieve perfect lab values but to improve the integrated hormonal milieu such that the client experiences better energy, recovery, performance, and body composition outcomes. The coach who understands endocrinology does not need to micromanage every variable, they identify the primary leverage point, intervene, monitor, and adjust. This is the application of hormonal primacy in practice.

Coaching Application

Domain 4 Case Study: The Complete Integration

Scenario: A 38-year-old male client, 92 kg, 22% body fat, has been training consistently for 2 years with moderate results. He reports low energy, reduced libido, poor sleep, and a stubborn midsection. He has never had blood work. His diet is adequate but unstructured; his training is 5x/week bodybuilding-style with inconsistent volume management. Stress is moderate-high from work and family.

Assessment: The integrated assessment reveals likely multi-axis dysfunction: elevated allostatic load (HPA axis), likely secondary hypogonadism (HPG axis) from visceral adiposity and chronic mild deficit, and possible low T3 syndrome. Labs are ordered to confirm.

Intervention: (1) Begin with HPA axis: stress management protocol, sleep optimisation, training volume reduction to 4x/week. (2) HPG axis: increase dietary fat to 0.8 g/kg, ensure adequate total calories, set body fat reduction as a medium-term goal to reduce aromatase burden. (3) HPT axis: transition from an unstructured eating pattern to a structured approach with adequate carbohydrate intake. (4) Initial labs reveal: total T 340 ng/dL, free T 52 pg/mL, cortisol (PM) 16 mcg/dL, TSH 2.1, free T3 2.8 (low-normal), SHBG 48 nmol/L. Confirms multi-axis dysfunction with HPA overdrive as the upstream driver.

Key Takeaway: The four-axis diagnostic framework identifies the primary dysfunction and directs the coaching intervention to the highest-leverage variable. When lifestyle factors are the root cause, no single supplement or training adjustment can substitute for addressing the upstream driver.

◈ MAKE THIS WORK FOR YOU

The integrated assessment is where the whole system comes together. Four layers exist because **no single marker tells the story**: subjective markers (libido, sleep, recovery, cycle, mood) catch what questionnaires measure poorly; anthropometrics (waist, body fat, muscle trends) catch the metabolic consequences; performance markers (strength, recovery rate, HRV) catch the output end; labs confirm and quantify. The coach's edge is pattern recognition – low libido + poor sleep + stalled strength + rising waist circumference identifies the axis before any blood test is drawn.

If you are coaching someone with hormonal symptoms: Run the layers in order – screen first (the Chapter 12 checklist), then subjective baseline, then anthropometrics and performance audit, and treat labs as confirmatory rather than primary. Follow the order of operations: rule out red flags (the functional vs pathological split from Chapter 20), address lifestyle first (energy availability, sleep, stress, training load), and only then interpret labs. When multiple axes are involved, address the HPA axis first – elevated cortisol suppresses both T3 and testosterone. Reassess on the 8–12 week loop; labs every second or third cycle.

If you are self-coaching: You can run the full assessment without a coach – the screening checklist, sleep quality, waist and body fat trends, and a strength log are all yours. Labs are the only layer requiring a professional. If symptoms persist after 8–12 weeks of a clean lifestyle (sleep, calories, training, stress), that is the moment for blood work – and the four-axis framework tells your clinician exactly which markers to order (Chapter 20's minimal panel).

The last rule of the book: The goal is not perfect lab values; it is better energy, recovery, performance, and body composition. Trends matter more than single snapshots, and a reference range is a population fact, not your target (Chapter 20). This assessment loop is what keeps the other 24 chapters honest – the hormonal playbook is complete.

Appendix A: Lab Reference Ranges and Interpretation

Table A.1 Complete Lab Reference Ranges for Hormonal Assessment

MARKER	MALE REFERENCE RANGE	FEMALE REFERENCE RANGE	OPTIMAL RANGE (FUNCTIONAL)
Total testosterone	250 to 1,100 ng/dL	15 to 70 ng/dL	500 to 900 (M), 30 to 60 (F)
Free testosterone	35 to 155 pg/mL	1 to 5 pg/mL	80 to 150 (M), 2 to 4 (F)
SHBG	10 to 57 nmol/L	18 to 144 nmol/L	20 to 40 (M), 40 to 80 (F)
LH	1.5 to 9.3 IU/L	2 to 15 IU/L (follicular)	3 to 8 (M), 3 to 10 (F)
FSH	1.4 to 18.1 IU/L	3 to 20 IU/L (follicular)	2 to 8 (M), 3 to 10 (F)
Prolactin	2 to 18 ng/mL	2 to 29 ng/mL	4 to 12
Estradiol (E2)	10 to 40 pg/mL	30 to 400 pg/mL (cycle-dependent)	20 to 30 (M), cycle-appropriate (F)
Cortisol (AM)	6 to 23 mcg/dL	6 to 23 mcg/dL	10 to 18
DHEA-S	80 to 560 mcg/dL	35 to 430 mcg/dL	200 to 400 (under 40)
TSH	0.4 to 4.5 mIU/L	0.4 to 4.5 mIU/L	1.0 to 2.5
Free T3	2.3 to 4.2 pg/mL	2.3 to 4.2 pg/mL	3.2 to 4.0
Free T4	0.8 to 1.8 ng/dL	0.8 to 1.8 ng/dL	1.0 to 1.5
Reverse T3	10 to 24 ng/dL	10 to 24 ng/dL	10 to 15
IGF-1	80 to 250 ng/mL (age-dependent)	80 to 250 ng/mL (age-dependent)	150 to 250 (under 40)
Ferritin	30 to 400 ng/mL	15 to 150 ng/mL	50 to 150
Vitamin D (25-OH)	30 to 100 ng/mL	30 to 100 ng/mL	50 to 80

Appendix B: Supplement Protocols Quick Reference

Table B.1 Supplement Protocols Quick Reference

SUPPLEMENT	DOSE	TIMING	CYCLE	PRIMARY INDICATION
Vitamin D3	1,000 to 3,000 IU	With largest meal (fat-soluble)	Continuous; recheck levels at 12 weeks	Low serum 25-OH-D; general hormonal support
Zinc (glycinate/picolinate)	15 to 30 mg	With evening meal	Continuous; ensure copper intake	Low T; low libido; poor recovery
Magnesium glycinate	200 to 400 mg	30 to 60 min before bed	Continuous	Poor sleep; stress; low T
Ashwagandha (KSM-66)	300 to 600 mg	With breakfast or split AM/PM	8 to 12 weeks then assess	Elevated cortisol; high perceived stress
Boron (glycinate/citrate)	3 to 6 mg	With breakfast	8 weeks then reassess	Low free T with elevated SHBG
D-Aspartic Acid	2,000 to 3,000 mg	Split AM/PM away from meals	12 to 28 days only	Short-term LH pulse augmentation
Creatine monohydrate	3 to 5 g	Post-training or with any meal	Continuous	Training performance; indirect hormonal support
Omega-3 (EPA/DHA)	1.5 to 4 g (EPA+DHA)	With largest meal	Continuous	Inflammation; general hormonal health

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